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Protostellar Multiplicity Survey in Taurus

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ABSTRACT

We present an ALMA (0.9 mm) and VLA (9 mm) survey of 27 protostars in the Taurus Molecular Cloud, comprising 3 Class 0, 23 Class I/FS, and 1 Class II objects observed at $0\farcs3$ ($\sim$20 au) resolution. To build a more complete sample of protostellar multiplicity in this region, we extended the sample with 24 additional protostars and their companions previously identified through infrared or ALMA observations. Within separations of 18–10,000 au, we measure a multiplicity fraction (MF) of 0.50 and a companion frequency (CF) of 0.58 for the observed sample, and an MF of 0.53 and CF of 0.72 for the extended sample. These values are notably higher than those reported in the more clustered star-forming regions of Orion and Perseus, though only at the $\sim$ 1–2σ level, suggesting that Taurus may preserve a larger fraction of primordial multiples. The separation distributions in our samples show populations of both close and wide multiples, but a deficit at intermediate separations of 200–300 au. This pattern suggests two distinct formation pathways: close binaries (<200 au) arising primarily from disk fragmentation, and wide multiples (>1000 au) from core fragmentation.

1. INTRODUCTION

Multiplicity is a common outcome of the star formation process, but the likelihood of a star having one or more companions depends strongly on its mass. The multiplicity fraction increases with stellar mass or spectral type; the most massive stars, O- and B-type, are almost always found in multiple systems, with multiplicity fractions around 90% (H. Sana et al. 2013, 2014; A. J. Frost et al. 2025). Solar-type stars (spectral types F, G, and K) have a multiplicity fraction of about 50% (D. Raghavan et al. 2010), while M-type (low-mass) stars are found in multiples roughly 30–35% of the time (C. J. Lada 2006). Some of these trends are seen consistently in both field populations and young clusters, at least for close multiples, suggesting that many of these multiple systems are formed early and survive into later evolutionary stages (J. Bouvier et al. 1997; J. Patience et al. 2002; P. Elliott et al. 2014; N. R. Deacon & A. L. Kraus 2020; G. Torres et al. 2021). Overall, the prevalence of multiples across stellar masses highlights the importance of studying multiplicity in the context of star formation.

To understand how multiple systems form and evolve, it is essential to observe their initial configurations during the earliest protostellar stages, when multiplicity is closest to its primordial state. Young stellar objects (YSOs) are classified into evolutionary stages based on quantities derived from their spectral energy distributions (SEDs). Class 0 sources are the youngest, deeply embedded in their natal envelopes and typically younger than 10^5 years (M. M. Dunham et al. 2014). Class I objects are slightly older (5x10^5 years), still embedded but with less dense envelopes. During these stages, circumstellar disks form due to angular momentum conservation during collapse (P. Cassen & A. Moosman 1981). Flat Spectrum protostars represent an intermediate phase where both disk and envelope contribute significantly to the emission, and Class II sources are largely free of envelopes, retaining only a protoplanetary disk (T. P. Greene et al. 1994). By the Class II phase, the separation distribution has already been reshaped and no longer cleanly traces the formation pathways.

Multiples are believed to form early in the star formation process, especially during the Class 0 stage, when the system still contains most of its mass reservoir. The two primary mechanisms are core fragmentation and disk fragmentation. Core fragmentation occurs at the scale of the core, typically forming wide multiples ($\gtrsim$1000 au) via turbulent fragmentation (S. S. R. Offner et al. 2010; A. T. Lee et al. 2019). In contrast, disk fragmentation involves gravitational instability in a rotationally supported disk, often producing closer multiples ($\lesssim$500 au; K. M. Kratter

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et al. (2010)). Over time, dynamical interactions and environmental effects reshape multiplicity: wide companions may migrate inward, be ejected, or be disrupted altogether, leading to reduced multiplicity at wide separations in older populations (S. S. R. Offner et al. 2023). It is important to note, the formation of close multiples remains an area of active debate, with some studies suggesting they may primarily originate from core fragmentation followed by substantial migration rather than disk fragmentation (S. S. R. Offner et al. 2023; S. I. Sadavoy & S. W. Stahler 2017; A. T. Lee et al. 2019; R. L. Kuruwita & T. Haugbølle 2023).

Direct observations of protostellar multiplicity are essential for testing theoretical models and understanding how systems evolve from their initial conditions. However, studying protostars has historically been difficult due to their deeply embedded nature. Fortunately, protostellar envelopes become optically thin at (sub)millimeter and centimeter wavelengths, making high-resolution observations with the Atacama Large Millimeter/submillimeter Array (ALMA) and the NSF's Karl G. Jansky Very Large Array (VLA) particularly well suited to detecting companions on scales below 1000 au, where mid-infrared instruments typically struggle to resolve them. These observations can penetrate the envelope and resolve the dense inner regions where multiple systems are likely to form.

Over the past decade, large interferometric surveys have enabled detailed multiplicity studies in two major star-forming regions: Perseus and Orion (J. J. Tobin et al. 2016, 2022). These surveys revealed a double-peaked separation distribution, with populations of both close and wide multiples–particularly for the youngest sources–broadly supporting the idea that multiple formation pathways (core and disk fragmentation) are at play on distinct scales. Among the three regions, Orion is the densest and the only one forming massive (OB) stars, with rich embedded clusters; Perseus is intermediate, combining clustered subregions with a more distributed population; and Taurus is the most diffuse and predominantly distributed, lacking massive stars. Studying multiplicity in Taurus therefore provides an important counterpoint, enabling us to explore how environment influences the formation and survival of multiple systems. To examine how multiplicity evolves over time, we also compare our findings to a survey of the more evolved Class II and III population in Taurus (A. L. Kraus et al. 2011).

In this study, we will characterize the occurrence rate and separation distribution of multiple protostars in the Taurus molecular cloud. We will then evaluate how they compare to those in other star-forming regions and more evolved Taurus populations, to better understand the formation mechanisms of multiple systems. Section 2 describes the observations and data analysis methodology, and Section 3 presents our observations. The multiplicity analysis is reported in Section 4, followed by a discussion of their implications in Section 5. Finally, Section 6 summarizes our conclusions.

2. OBSERVATIONS & DATA ANALYSIS

2.1. *ALMA Observations*

We observed 27 protostars in the Taurus star-forming region with ALMA in Cycle 7 (project 2019.1.00847.S) using the 12 m array in Band 7 (340 GHz; 0.9 mm). The observations were conducted from 2019 to 2022 across three scheduling blocks (SBs) grouped by source location, all using the C43-4 configuration, which provided baselines from 15 m to 784 m. The corresponding angular resolution was $\sim0\farcs3$, and the largest angular scale was $\sim4''$. The continuum sensitivity was typically $\sim$0.29 mJy/beam. Block 1 (2019 October 7) consisted of two executions covering five sources at $0\farcs265$ resolution, block 2 (2019 October 7) contained one execution for a single source at $0\farcs309$ resolution, and block 3 (2022 August 8) comprised six executions targeting 21 sources at $0\farcs208$ resolution. The phase calibrators, number of antennas, and precipitable water vapor (PWV) levels varied between executions and are listed in Table 1. On-source integration times ranged from 13.3 to 22.7 minutes per field and are provided in Table 2, which summarizes additional observing parameters by source.

The correlator was configured with four basebands. One provided a continuum window centered at 328.750 GHz with 1.875 GHz total bandwidth in 31.25 MHz channels. A second baseband targeted the ^{13}CO and C^{18}O $J=3\rightarrow2$ lines with two spectral windows centered at 330.588 and 329.331 GHz, respectively; each spanned 234.38 MHz and delivered velocity resolutions of 0.256 and 0.257 km s^{-1}. A third baseband had four spectral windows covering CN ($N=3\rightarrow2$) at 339.447, 339.517, 340.032, and 340.248 GHz, each 117.19 MHz wide with 0.249 km s^{-1} channels. The fourth baseband was centered on CS ($J=7\rightarrow6$) at 342.883 GHz with 468.75 MHz bandwidth and 0.247 km s^{-1} channels. Line-free channels from the line spectral windows were combined with the dedicated continuum spectral window, yielding an aggregate continuum bandwidth of $\simeq$3.0 GHz. For the purpose of this paper, we do not discuss the molecular line data and instead focus solely on the continuum emission.

The ALMA data taken in 2019 were calibrated with the CASA pipeline version 5.6.1-8, while the 2022 data were calibrated with pipeline version 6.2.1-7. The ALMA observations have an absolute flux calibration uncertainty of approximately 10%, but only statistical uncertainties are considered in our analysis.

2.2. *VLA Observations*

We observed the same 27 protostars with the VLA under project 18B-179. All sources were observed in B configuration between 2019 June 3 and June 26, with a subset of 14 also observed in BnA configuration between 2019 June 28 and July 17. The B configuration has a maximum baseline of 11.1 km, yielding a resolution of $0.''19$ (26.6 au) at 9 mm, while the hybrid BnA configuration extends the maximum baseline along the north arm of the array to 22.8 km, achieving a resolution of $0.''15$ (21 au) in that direction.

All observations were conducted in Ka-band using the 3-bit samplers in 8 GHz continuum mode, with one 4 GHz baseband centered at 35.0 GHz and the other at 31.0 GHz. The full 8 GHz bandwidth was divided into 128 MHz spectral windows, each containing 64 channels with 2 MHz resolution. On-source integration times ranged from approximately 7 to 14 minutes and are listed in Table 2. 3C147 served as the flux calibrator, and J0319+4130 as the bandpass calibrator across all observations. The specific phase calibrator varied between scheduling blocks and is also listed in Table 2. We alternated between target and phase calibrator scans, spending roughly 80-100 s on the calibrator after each $\sim$100 s scan of a science target.

The calibration for VLA observations obtained on or before 2019-06-17 used version 5.4.1-32 of the VLA pipeline, while observations taken after that date used version 5.4.2-5. The only exception was the archive file 18B-179.sb36806445.eb36822423.58662.55088293982, which failed in the original pipeline and we subsequently reprocessed with the VLA pipeline version 6.5.4-9 (CASA Team et al. 2022). The VLA data likewise carry an absolute flux calibration uncertainty of roughly 10%, though only statistical uncertainties are included in our analysis.

2.3. *Self Calibration & Imaging*

To the pipeline-calibrated data, we then applied the self-calibration pipeline developed by J. Tobin and P. Sheehan[3] to both the ALMA and VLA data, with additional manual flagging applied where necessary. Table 2 summarizes which observations were successfully self-calibrated. Imaging was performed with the `tclean` task in CASA using Briggs weighting. For ALMA we imaged each target with robust parameters $R = -1.0$, 0.5, 0.0, 1.0; for the VLA we used $R = 0.5$, 1.0, or 2.0 depending on source. All ALMA images cover $31'' \times 31''$. To maintain adequate sampling of the synthesized beam, we adopted a cell size of $0.0155''\,\mathrm{pix}^{-1}$ for the higher–resolution weightings ($R = -1.0$, 0.5; 2000×2000 pixels) and $0.031''\,\mathrm{pix}^{-1}$ for the lower–resolution weightings ($R = 0.0$, 1.0; 1000×1000 pixels). VLA images were made with a cell size of $0.020''\,\mathrm{pix}^{-1}$, yielding a $122'' \times 122''$ field of view (6144×6144 pixels).

Typical synthesized beams were $0.28'' \times 0.20''$ for ALMA and $0.33'' \times 0.18''$ for the VLA (FWHM; varying with robust and uv coverage), corresponding to $\sim 39 \times 28$ au and $\sim 46 \times 25$ au at 140 pc. The median continuum rms was $\sim$0.66 mJy beam^{-1} at 0.9 mm and $\sim$0.020 mJy beam^{-1} at 9 mm.

2.4. *Archival Data*

We include two previously unpublished archival datasets from the NRAO Science Data Archive. The observations and data reduction procedures for these datasets are described as follows.

IRAS 04295+2251 was observed by the VLA in Ka-band in B-configuration on 2015 February 19 and during the A-configuration to D-configuration move on 2015 September 25 (Project 15A-381). The correlator was configuration for full 3-bit continuum mode, with one baseband centered at $\sim$29 GHz and the other centered at $\sim$37 GHz, sampling 8 GHz of bandwidth, with a 4 GHz gap between the basebands; C-band data were also observed as part of the same EB, but those data will not be discussed here. The observed basebands were divided into 64, 128 MHz wide spectral windows, each with 64 channels. Approximately 50 minutes were spent on source, after slew overheads in both observations. For both observations, J0440+2728 was observed as the complex gain calibrator, 3C84 was observed as the bandpass calibrator, and 3C147 was observed as the flux density calibrator. Both the A and B-configuration data were originally calibrated using the scripted VLA pipeline using CASA 4.2.2. The observations conducted during the A to D configuration move had 18 antennas still in the A-configuration positions and the other 9 antennas had already been moved to their D-configuration positions. All antennas were used during calibration of that dataset, but

[3] https://github.com/jjtobin/auto_selfcal

Table 1. ALMA Observing Details by Execution Block

MOUS[a]	EB ID[b]	Date	N_{ant}	PWV (mm)	ALMA Phase Calibrator
uid://A001/X1467/X260	uid://A002/Xe1d2cb/Xc90c	2019-10-07	45	0.46	J0431+1731
”	uid://A002/Xe1d2cb/Xd227	2019-10-07	45	0.41	J0431+1731
uid://A001/X1467/X267	uid://A002/Xe1d2cb/Xda79	2019-10-07	45	0.44	J0438+3004
uid://A001/X1467/X26e	uid://A002/Xfc69ac/Xa3ed	2022-08-05	46	0.40	J0438+3004
”	uid://A002/Xfaf3de/X2fd6	2022-08-05	45	0.33	J0438+3004
”	uid://A002/Xfaf3de/X2af2	2022-08-05	45	0.32	J0438+3004
”	uid://A002/Xfa5545/X397f	2022-08-05	46	0.50	J0438+3004
”	uid://A002/Xfa2f45/X18c6d	2022-08-05	41	0.46	J0438+3004
”	uid://A002/Xfa0be4/Xfda0	2022-08-05	41	0.37	J0438+3004

Note—[a]Member Observation Unit Set; [b]Execution Block. Together, the MOUS and EB ID provide unique identifiers assigned by ALMA.

prior to imaging, we flagged all the visibilities with uv-distances < 100 kλ in order to prevent the short baselines from creating a poor beam. The data were self-calibrated using the automated self-calibration routines as used on the other ALMA and VLA observations mentioned previously. We first ran the self-calibration on the B-configuration data alone, then we copied the self-calibrated B-configuration data into a new directory and self-calibrated the A-configuration data together with the previously self-calibrated B-configuration data. This yields a superior result because the B-configuration data alone could be self-calibrated to a solution interval that was the length of a single scan, while the self-calibration of the A- and B-configuration together could only be self-calibrated on a solution interval spanning an the entire EB. This is at least in part due to the A-configuration data only have 18 usable antennas for self-calibration.

The proper motion of IRAS 04295+2251 has caused the apparent source position to move relative to the position at the time of the ALMA observations, from the VLA observations being conducted in 2015. Thus, we need to compute the proper motion of the source in order to properly align the coordinates of the source with the ALMA data. To compute the proper motion, we use the VLA data taken in B-configuration on 2015 February 19 (MJD 57072) and the data taken on 2019 June 18 (MJD 58652) to calculate the proper motion of IRAS 04295+2251. The position on MJD 57072 was α=04:32:32.0785 δ=22:57:26.204 and on MJD 59652 was α=04:32:32.0801 δ22:57:26.158, which were determined using the CASA task imfit to fit Gaussians to the unresolved point source in the images. The proper motion is found to be μ_α=5.25 mas yr^{-1} and μ_δ=-10.59 mas yr^{-1}. The ALMA observations were taken approximately on 2022 July 2 MJD 59762 and the total shift in position between the VLA data taken in 2015 is 38.67 mas in right ascension and -77.98 mas in declination. We used these values to adjust the reference positions for the image headers of the 2015 VLA data in order to bring their coordinates into agreement with those of the ALMA position from the observations in 2022.

IRAM 04191 was observed with ALMA Band 6 (project 2016.1.01284.S) using the 12 m array in two executions with different configurations. The first observation was obtained on 2016 October 15 in configuration C40-6, providing baselines from 15 to 1800 m, while the second was taken on 2017 July 21 in configuration C40-5, spanning 17-1100 m. Together, the data yield a synthesized beam of $0\farcs154$ and a largest angular scale (LAS) of 1.95″, with a continuum sensitivity of 0.017 mJy/beam and a field of view of 25.6″. The 2016 execution used J0433+0521 as the phase calibrator, with 41 antennas operational and precipitable water vapor (PWV) of 0.64 mm. The 2017 execution used J0449+1121 as the phase calibrator, with 46 antennas and similar atmospheric conditions (PWV = 0.64 mm). The total on-source integration time for the combined dataset is 3992 s. The correlator was configured with four basebands. One baseband contained four high-resolution spectral windows centered at 218.436, 218.459, 219.558, and 219.948 GHz. The remaining three basebands provided continuum coverage centered at 221.36, 233.50, and 235.50 GHz, each 2.0 GHz wide with 977 kHz channels, yielding an effective continuum bandwidth of $\sim$6 GHz.

We applied the same self-calibration procedure described in Section 2.3. No proper-motion correction was performed, as the position of the source identified as IRAM 04191-A (also detected in our original ALMA dataset) showed only a negligible positional shift between observing runs.

Table 2. Observing Parameters Per Source

Source	ALMA (0.9 mm)		VLA (9 mm)		
	T.O.S.	Self Cal.	T.O.S.	Self Cal.	Phase Cal.
04016+2610	798s	✓	717s	✗	J0403+2600
04108+2803	871s	✓	789s	✓	J0403+2600
04158+2805	871s	✓	813s	✗	J0403+2600
04169+2702	798s	✓	688s	✓	J0429+2724
04181+2654 A	798s	✓	688s	✗	J0429+2724
04166+2706	798s	✓	688s	✓	J0429+2724
04381+2540	798s	✓	806s	✗	J0429+2724
DG Tau	798s	✓	782s	✓	J0429+2724
04260+2642	798s	✓	751s	✗	J0429+2724
HH_30	968s	✓	535s	✗	J0431+1731
L1551NE	968s	✓	419s	✓	J0431+1731
04287+1801	968s	✓	502s	✓	J0431+1731
04191+1523	907s	✓	551s	✗	J0431+1731
IRAM04191	907s	✓	551s	✗	J0431+1731
04264+2433	798s	✓	782s	✓	J0426+2327
04302+2247	798s	✓	766s	✗	J0426+2327
04295+2251	798s	✓	766s	✓	J0426+2327
04263+2426	798s	✓	790s	✓	J0426+2327
04489+3042	1361s	✓	750s	✗	J0438+3004
04385+2550	798s	✓	774s	✗	J0438+3004
04325+2402	798s	✓	750s	✗	J0438+3004
04365+2535	798s	✓	582s	✓	J0429+2724
04368+2557	798s	✓	588s	✓	J0429+2724
04361+2547	798s	✓	588s	✓	J0429+2724
04181+2654 B	798s	✗	782s	✗	J0429+2724
04239+2436	798s	✓	798s	✓	J0429+2724
04248+2612	798s	✗	766s	✗	J0429+2724

Note—Because of slew overheads, the actual VLA TOS is ∼30% less than the values reported in this table. Check marks (✓) indicate observations that were successfully self-calibrated; crosses (✗) indicate those that were not.

2.5. *The Taurus Sample*

The Taurus Molecular Cloud is a nearby star-forming region, at an average distance of ∼140 pc. Unlike more active regions such as Orion or Perseus, Taurus is characterized by a low stellar density (L. M. Rebull et al. 2010) and a somewhat older protostellar population, containing only four known Class 0 protostars compared to 40 Class I and Flat Spectrum sources. The low stellar density in Taurus suggests it may offer insight into multiplicity in environments with minimal dynamical interactions. This provides a contrasting environment to test whether trends seen in more clustered regions hold in regions like Taurus. The original sample for this study consisted of 3 Class 0, 23 Class I/FS, and 1 Class II protostars. These sources were selected based on consistent classifications across multiple studies (e.g., F. Motte & P. André (2001); S. M. Andrews & J. P. Williams (2005); E. Furlan et al. (2008)). The full source catalog for these observations is presented in Table 3, which lists the classes, coordinates, distances, bolometric luminosities ($L_{\rm bol}$), bolometric temperatures ($T_{\rm bol}$), and ALMA and VLA fluxes and rms values for all components in each system.

Table 3. Taurus Source Catalog: Observed Sample

Source	Cl.[a]	R.A.	Decl.	$L_{\rm bol}$	$T_{\rm bol}$	α[b]	Dist.	#C[c]	**ALMA (0.9 mm)**		**VLA (9 mm)**	
		(J2000)	(J2000)	($L_\odot$)	(K)		(pc)		$\boldsymbol{I_\nu}$, Peak[d] (mJy)	rms (mJy)	$\boldsymbol{I_\nu}$, Peak[d] (mJy)	rms (mJy)
IRAS 04016+2610	I	4:04:43.08	+26:18:56.11	3.84	204	0.98	149.0	1	13.78	0.18	0.19	0.021
IRAS 04108+2803-B	I	4:13:54.73	+28:11:32.25	0.52	203	0.95	130.0	2	50.82	0.35	0.39	0.021
IRAS 04158+2805-A	I	4:18:58.14	+28:12:22.70	0.14	427	0.33	130.0	2	5.14	0.24	0.06	0.019
IRAS 04158+2805-B	I	4:18:58.15	+28:12:22.78	0.14	427	0.33	130.0	2	3.98	0.24	...	0.019
IRAS 04166+2706	0	4:19:42.51	+27:13:35.79	0.45	56	2.70	158.0	1	107.27	0.70	1.07	0.019
IRAS 04169+2702	I	4:19:58.48	+27:09:56.80	1.64	161	1.33	158.0	1	123.16	0.90	0.52	0.019
IRAS 04181+2654-B	I	4:21:10.39	+27:01:37.27	0.30	306	0.33	160.0	2	...	0.10	...	0.021
IRAS 04181+2654-A	I	4:21:11.49	+27:01:08.95	0.73	252	0.43	158.0	2	12.13	0.13	0.17	0.021
IRAM 04191-A	0	4:21:56.90	+15:29:46.05	0.10	28	1.61	144.0	4	7.21	0.11	...	0.020
IRAS 04191+1523-A	I	4:22:00.43	+15:30:21.18	0.52	89	1.03	144.0	4	80.95	0.75	0.16	0.022
IRAS 04191+1523-B	I	4:22:00.09	+15:30:24.59	0.52	89	1.03	144.0	4	31.39	0.75	...	0.022
IRAS 04239+2436-A	I	4:26:56.26	+24:43:34.76	1.27	257	0.81	128.0	2	43.86	0.37	0.64	0.040
IRAS 04239+2436-B	I	4:26:56.28	+24:43:34.75	1.27	257	0.81	128.0	2	42.23	0.37	0.55	0.040
IRAS 04248+2612-A	I	4:27:57.34	+26:19:17.89	0.33	224	0.52	128.0	3	2.57	0.10	...	0.022
IRAS 04248+2612-B	I	4:27:57.32	+26:19:17.78	0.33	224	0.52	128.0	3	2.30	0.10	...	0.022
IRAS 04248+2612-C	I	4:27:56.37	+26:19:17.61	0.33	224	0.52	128.0	3	1.15	0.10	...	0.022
IRAS 04260+2642	I	4:29:05.00	+26:49:06.79	0.08	353	0.31	128.0	1	30.84	0.39	0.17	0.019
IRAS 04263+2426-A	I	4:29:23.73	+24:33:01.01	7.66	351	0.87	128.0	2	86.97	0.62	0.83	0.017
IRAS 04263+2426-B	I	4:29:23.75	+24:32:59.66	7.66	351	0.87	128.0	2	65.40	0.62	1.53	0.017
IRAS 04264+2433-A	I	4:29:30.10	+24:39:54.54	0.39	209	0.96	128.0	2	16.22	0.16	0.25	0.017
IRAS 04264+2433-B	I	4:29:30.10	+24:39:54.93	0.39	209	0.96	128.0	2	4.20	0.16	0.06	0.017
IRAS 04287+1801-A	I	4:31:34.16	+18:08:04.68	26.79	111	1.91	144.0	2	322.49	2.37	3.03	0.023
IRAS 04287+1801-B	I	4:31:34.17	+18:08:04.26	26.79	111	1.91	144.0	2	191.53	2.37	2.29	0.023
L1551 NE-A	I	4:31:44.51	+18:08:31.33	3.18	101	0.90	144.0	2	242.49	2.32	2.09	0.022
L1551 NE-B	I	4:31:44.48	+18:08:31.57	3.18	101	0.90	144.0	2	97.87	2.32	0.87	0.022
IRAS 04295+2251 AB	I	4:32:32.08	+22:57:26.10	0.68	337	0.44	161.0	2	9.49	0.42	0.56	0.016
IRAS 04302+2247	FS	4:33:16.50	+22:53:20.22	0.41	181	0.12	161.0	1	29.15	0.70	0.15	0.016
IRAS 04325+2402-A	I	4:35:35.42	+24:08:18.78	0.85	112	0.91	128.0	2	15.65	0.27	0.20	0.019
IRAS 04325+2402-B	I	4:35:35.32	+24:08:26.79	1.55	77	1.70	128.0	2	11.18	0.27	0.09	0.019
IRAS 04361+2547-A	I	4:39:13.91	+25:53:20.33	3.30	138	1.56	140.0	2	71.70	0.54	1.12	0.020
IRAS 04365+2535	I	4:39:35.21	+25:41:44.08	2.45	183	1.19	140.0	1	151.81	1.25	1.20	0.021
IRAS 04368+2557	0	4:39:53.88	+26:03:09.40	1.50	40	2.93	140.0	1	100.71	0.96	1.04	0.021
IRAS 04381+2540-A	I	4:41:12.69	+25:46:34.63	0.62	174	0.97	140.0	2	25.90	0.20	0.32	0.019
IRAS 04381+2540-B	I	4:41:12.73	+25:46:34.65	0.62	174	0.97	140.0	2	13.56	0.20	0.24	0.019
IRAS 04385+2550	FS	4:41:38.84	+25:56:26.30	0.45	607	-0.04	140.0	1	42.16	0.28	0.18	0.019
IRAS 04489+3042-A	I	4:52:06.69	+30:47:16.93	0.39	424	0.39	156.0	2	22.46	0.22	...	0.017
IRAS 04489+3042-B	I	4:52:06.74	+30:47:19.83	0.39	424	0.39	156.0	2	0.54	0.22	...	0.017
DG Tau-B	I	4:27:02.58	+26:05:30.08	0.91	194	1.29	128.0	2	159.96	1.71	1.40	0.017
DG Tau-A	FS	4:27:04.70	+26:06:15.66	3.58	632	0.19	128.0	2	...	1.71	0.54	0.017
HH 30	II	4:31:37.49	+18:12:23.76	0.02	320	-0.31	144.0	1	4.79	0.15	...	0.020

NOTE— The entries where the peak flux is absent were the non-detections, with only two such instances in the ALMA data, and ten in the VLA data. All intensities and rms values are reported per beam. Superscripts indicate: [a]Evolutionary class. [b]Infrared spectral index. [c]Number of components in the system, which include companions from the Taurus+ sample. [d]Peak specific intensity.

Initially, Flat Spectrum (FS) sources were excluded from the sample, along with several Class I sources previously identified as FS. However, to ensure completeness for this multiplicity survey, the sample was later expanded to include these sources, even where classifications in the literature were inconsistent, which we hereafter refer to as the Taurus+ sample. Most of these sources have previously been observed at infrared or radio wavelengths with resolutions comparable to this survey, but did not typically have archival ALMA or VLA data. This extension added 2 Class 0, 9 Class I, and 8 FS sources. Additionally, we included 4 Class II sources and 1 Class III source that are components of systems containing sample members. Table 4 provides the full list of these extended sources, along with their properties, and you can find individual descriptions and references from the literature for each in Appendix A.2.

Table 4. Taurus Source Catalog: Extended Sample

Source	Class	R.A. (J2000)	Decl. (J2000)	L_{bol} ($L_\odot$)	T_{bol} (K)	α[a]	Distance (pc)	# Components
IRAS 04108+2803-A	FS	4:13:53.40	+28:11:22.92	0.14	731	-0.26	130.0	2
IRAS 04154+2823	FS	4:18:32.05	+28:31:14.99	0.38	517	0.27	130.0	1
IRAS 04166+2708	I	4:19:41.53	+27:16:06.67	0.13	170	0.68	158.0	1
2MASS J04194657+2712552	FS	4:19:46.57	+27:12:55.22	0.05	541	0.14	158.0	1
IRAS 04181+2655	I	4:21:07.97	+27:02:20.08	0.52	378	0.51	160.0	1
IRAS 04181+2654-B	I	4:21:10.39	+27:01:37.27	0.30	306	0.33	160.0	2
IRAM 04191-B	0	4:21:57.67	+15:29:51.07	0.10	28	1.61	144.0	4
FS Tau-B	I	4:22:01.01	+26:57:35.28	0.58	174	1.18	140.0	2
FS Tau-A	FS	4:22:02.20	+26:57:30.38	3.39	240	-0.22	140.0	2
L1521F	0	4:28:38.96	+26:51:34.99	0.36	29	1.43	140.0	1
2MASS J04293209+2430597	I	4:29:32.09	+24:30:59.76	0.06	360	0.83	128.0	1
LkHα358	FS	4:31:36.14	+18:13:43.21	0.20	875	-0.16	144.0	3
HL Tau	I	4:31:38.51	+18:13:57.86	8.69	307	0.53	147.3	3
XZ Tau	II	4:31:40.08	+18:13:56.64	1.83	1362	-1.52	147.3	3
Haro 6-13	FS	4:32:15.42	+24:28:59.59	1.29	654	-0.03	128.6	1
IRAS 04295+2251-B	I	4:32:32.07	+22:57:26.15	0.68	337	0.44	161.0	2
HP Tau	FS	4:35:52.77	+22:54:23.15	1.40	912	-0.24	176.3	5
HP Tau G3	II	4:35:53.38	+22:54:09.89	0.32	1285	-0.91	176.0	5
HP Tau G2	III	4:35:54.00	+22:54:13.46	1.27	2233	-1.69	176.0	5
Haro 6-28 B	II	4:35:56.82	+22:54:35.53	0.17	1591	-0.90	162.5	5
Haro 6-28 A	II	4:35:56.87	+22:54:35.82	0.17	1591	-0.90	162.5	5
IRAS S04361+2331	FS	4:39:05.26	+23:37:44.76	0.07	601	0.19	161.0	1
IRAS 04361+2547-B	I	4:39:13.90	+25:53:20.13	3.30	138	1.56	140.0	2
IC 2087 IR	I	4:39:55.73	+25:45:01.73	3.26	573	0.84	135.0	1

NOTE— [a]Infrared spectral index.

To refine distances to individual sources, we cross-matched our target coordinates with the stellar groups identified by K. L. Luhman (2023) using Gaia data. This yields more accurate distances spanning 130–160 pc, which are used throughout our analysis. A map of the Taurus Molecular cloud and locations of our sources therein can be found in Figure 1. The background extinction map is based on the dust map from D. J. Schlegel et al. (1998), which provides full-sky estimates of interstellar reddening E(B–V). The map was constructed from a combination of infrared observations by IRAS and COBE/DIRBE, tracing thermal emission from interstellar dust. We use the implementation provided by the dustmaps Python package described in G. Green (2018).

2.6. *The Orion & Perseus Sample*

The comparison samples used in this study come from the VLA/ALMA Nascent Disk and Multiplicity (VANDAM) surveys of the Perseus and Orion molecular clouds (J. J. Tobin et al. 2016, 2022). These regions represent high-density, actively star-forming environments that differ from the more quiescent Taurus region. The first of these surveys was conducted in 2016 towards Perseus at a distance of $\sim$300 pc. They used the VLA at 9 mm at a resolution of $0\farcs07$ and were able to resolve companions as small as 24 au separation. This work observed 104 systems in total: 55 Class 0 protostars, 37 Class I and FS protostars, and 12 Class II systems. These sources were primarily selected from M. L. Enoch et al. (2009), and span a bolometric luminosity range of $\sim$0.1 to 120 $L_\odot$.

The second survey was toward Orion at a distance of $\sim$400 pc and contains a large and diverse protostellar population. The VANDAM Orion sample includes 94 Class 0 protostars, 128 Class I protostars, and 103 Flat Spectrum (FS) sources, for a total of 325 sources drawn primarily from the Herschel Orion Protostar Survey (HOPS) (E. Furlan et al. 2016). They observed these sources with ALMA at 0.87 mm at an angular resolution at $0\farcs1$ and observed a subset of these sources with the VLA at 9.1 mm and $0\farcs07$ resolution. With these observations, the smallest companion separation they were able to resolve was 21.8 au. The bolometric luminosities for these sources span a wide range, from $\sim$0.1 to $\sim$ 1400 $L_\odot$.

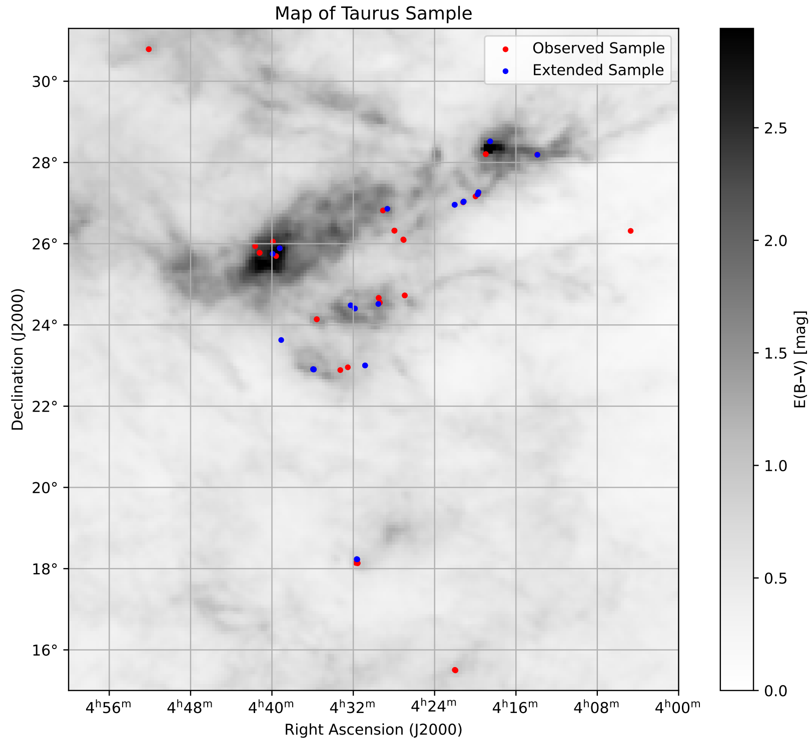

Figure 1. Map of the Taurus Molecular Cloud showing the locations of the sources overlaid on a background extinction map from D. J. Schlegel et al. (1998). Sources from the original sample are shown in red, and additional sources included in the Taurus+ sample are shown in blue.

Both Orion and Perseus host significantly larger young protostellar populations, with a higher proportion of Class 0 sources, and more massive stars forming within dense clusters. Because of the relatively high YSO density, they applied a correction for chance alignments when identifying companions. This was done by calculating a weighted probability for each candidate companion based on the local surface density of YSOs, the projected separation between sources, and the statistical likelihood of random alignment. We will proceed referencing the contamination corrected results for our comparisons. No such correction was applied to the Taurus sample due to its significantly lower source density, which reduces the chance of spurious associations. These datasets provide a benchmark for understanding multiplicity in denser, more massive star-forming regions and serve as a useful contrast to our Taurus sample.

2.7. *More Evolved Taurus Sample*

While Orion and Perseus allow us to compare Taurus to other star-forming regions, it is also important to examine how multiplicity evolves within the same environment over time. To this end, we compare our results to a sample of more evolved Class II and III stars in Taurus from A. L. Kraus et al. (2011). This provides a direct view of how the multiplicity properties of protostars may change as systems age.

A. L. Kraus et al. (2011) conducted a high-resolution multiplicity survey of Class II/III stars in the Taurus Molecular Cloud using the Keck and Palomar telescopes. Their sample included 128 sources and was sensitive to separations ranging from 3 to 5000 au, meaning that their survey did not cover the widest multiples probed in our study.

A key aspect of their analysis was to investigate how multiplicity varies with stellar mass. Stellar masses were estimated from spectral types, with uncertainties of up to 20%. Their full sample spanned 0.25–2.5$M_{\odot}$, and they divided it into subsamples of low-mass and high-mass primaries. They found that lower-mass stars showed a notable lack of companions at separations above approximately 200 au. Overall, their findings supported the idea that binary formation operates through two distinct mechanisms: large-scale fragmentation of collapsing cores leading to wide multiples, and disk fragmentation producing closer companions.

2.8. *Data Analysis*

Source identification and flux measurements were performed using the CASA `imfit` task, which fits two-dimensional elliptical Gaussians to the continuum emission. For some systems with complex or blended structures, multiple Gaussians were fit to better capture the morphology. To characterize multiplicity across the sample, we created a merged source catalog incorporating detections from both ALMA and VLA. This approach helps account for non-detections from either observatory and ensures a more complete census of companions. We also assembled photometry and fit SEDs for each target to derive bolometric quantities and evolutionary classes. A full description of this process and the resulting SEDs is provided in Appendix B.

2.8.1. *Associating Multiple Systems*

To study multiplicity, we must first identify which sources are physically associated as part of the same system. We do this using spatial association techniques that group sources based on their projected separations on the sky. We adopt the iterative "modified nearest neighbor" method introduced in J. J. Tobin et al. (2022), which builds multiple systems by first identifying the closest pair of sources within a minimum threshold of 15 au, which is smaller than the separation of any source in our sample, chosen conservatively to ensure no close pairs are missed. When a pair is found, the algorithm computes their geometric average position, replaces the pair with that average, and repeats the search for additional companions relative to the new location. The original sources are removed from the catalog after each association to ensure that no source is linked to more than one system. This procedure is performed across 100 logarithmically spaced radial bins, with a maximum separation of 10,000 au. We adopt 10,000 au as our maximum separation limit because it approximately corresponds to the typical radius of dense cores in which protostars form (P. J. Benson & P. C. Myers 1989; E. A. Bergin & M. Tafalla 2007; J. Lane et al. 2016; H. Kirk et al. 2017).

A key advantage of this method over more traditional approaches, which typically define a primary source and apply a fixed radial cutoff, is that it can recover more complex and extended systems. This method also avoids the need to designate one component of a system as the "primary." Such designations are problematic, as the brightest or most luminous protostar at infrared, millimeter, and/or centimeter wavelengths is not always the most massive. Protostellar luminosity can be dominated by accretion processes rather than intrinsic stellar properties, and dust continuum emission (our main observable) does not directly correlate with stellar mass (M. M. Dunham et al. 2014; W. J. Fischer et al. 2017).

2.8.2. *Multiplicity Statistics*

To characterize the multiplicity of protostars in Taurus, we use two main metrics: the multiplicity fraction (MF) and the companion frequency (CF).

The MF is the fraction of systems that are part of multiple systems (i.e., have at least one companion), and is calculated by adding up the number of binaries (B), triples (T), quadruples (Q), etc, and dividing by the total number of systems ($N_{\rm sys}$), including the singles as given by

$$MF = \frac{B + T + Q + \ldots}{S + B + T + Q + \ldots}. \tag{1}$$

The CF, which represents the number of companions each source has on average, is given by

$$CF = \frac{B + 2T + 3Q + \ldots}{S + B + T + Q + \ldots}. \tag{2}$$

To estimate the uncertainty in this analysis, we compute Wilson score intervals (E. B. Wilson 1927), a binomial statistics method, for all MFs and most CFs across different separation ranges, given by

$$\sigma_{\rm MF} = \frac{1}{1 + \frac{z^2}{N_{\rm sys}}} \left({\rm MF} + \frac{z^2}{2N_{\rm sys}} \right) \pm \frac{z}{1 + \frac{z^2}{N_{\rm sys}}} \sqrt{\frac{MF\,(1.0 - MF)}{N_{\rm sys}} + \frac{z^2}{4N_{\rm sys}^2}}. \tag{3}$$

This shows the formula for calculating the uncertainty in MF and can be applied analogously to CF by replacing MF with CF. Using z=1.0 yields a 68% confidence interval, consistent with 1-sigma uncertainties. This method provides asymmetric confidence intervals for binomial proportions, which is especially important when the number of systems ($N_{\rm sys}$) is small or the proportion is near 0 or 1, where normal approximations become unreliable. However, for CF values approaching or exceeding 1.0, the Wilson interval becomes inaccurate and can even produce non-physical results (e.g., negative values under the square root). This occurs because the CF, unlike the MF, can exceed 1.0 when systems have multiple companions. In such cases, we instead adopt Poisson statistics to estimate the uncertainty, using the standard deviation of the number of companions. This approach is more appropriate when CF>0.5, as it treats the companion count as a Poisson-distributed quantity rather than a binomial proportion.

2.8.3. *Comparing Multiplicity Properties*

MF and CF offer a general overview of how common multiples are in a given region. These statistics can be calculated for all systems or within specified separation ranges to examine how multiplicity varies with separation. Since protostellar systems may contain a mix of evolutionary classes, especially at wider separations, we include certain Class II sources when they are confirmed companions to Class 0, I, or Flat Spectrum protostars. However, for simplicity and statistical robustness, we do not break down MF and CF by individual class. Instead, we assign each system a single class based on the majority class present. If the number of components from different classes is equal, the system inherits the earlier class.

To explore more detailed trends, we analyze the distribution of companion separations using histograms and cumulative distribution functions (CDFs). These allow us to visually and statistically compare different regions. CDFs are particularly useful for revealing how companions are distributed across a range of separations: steep slopes indicate that companions are clustered at smaller separations, while flatter slopes suggest a more even spread.

In Taurus, we construct standard CDFs since no correction for background YSO density is applied. A CDF is a curve that starts at 0 and rises to 1, where the x-axis is projected separation (in au), and the y-axis represents the fraction of systems with separations less than or equal to a given value. The empirical CDF is defined as

$$\mathrm{CDF}(d_n) = \frac{n}{N}, \quad \text{for } n = 1, 2, \ldots, N, \tag{4}$$

where N is the total number of companion systems and d_n is the nth smallest projected separation.

To quantitatively compare separation distributions between regions, we use the Kolmogorov–Smirnov (KS) test and the Anderson-Darling (AD) test on the CDFs of companion separations, and we report results from both tests. The KS test measures the maximum difference between two cumulative distributions and is most sensitive around the median, while the AD test gives more weight to the distribution tails. We compare the Taurus and Taurus+ samples with Orion, Perseus, and the more evolved Taurus protostars (low- and high-mass) from A. L. Kraus et al. (2011), as well as with benchmark distributions including a log-flat model and the solar-type field stars from D. Raghavan et al. (2010). These statistical tests evaluate the null hypothesis that two samples are drawn from the same parent population. If the resulting p-value is below 0.01, we reject the null hypothesis and conclude that the distributions are statistically distinct.

While MF/CF statistics and CDFs both provide insight into multiplicity, they probe different aspects of the population and can yield different conclusions. For example, two regions might have similar MF and CF values but different shaped separation distributions—or vice versa. Therefore, using both approaches allows for a more complete understanding of how protostellar multiplicity varies across environments.

3. RESULTS

The sources were observed in continuum with ALMA at 0.9 mm and the VLA at 9 mm. The observed continuum maps from our primary observations, with close-up views, are shown in Figures 2 (ALMA) and 3 (VLA), while Figure 4 presents wider-field views of the most widely separated multiple systems. All of these companions were previously detailed in literature at infrared wavelengths except for the close companion to IRAS 04489+3042.

We present the multiple systems observed in our Taurus sample in Table 5, including their classifications and projected separations. In systems with more than two components, the listed separation to a third component or sub-system refers to the distance from the geometric center of the first pair. This allows for a consistent measure of separation across multi-source systems.

Within our observed sample, we detect 12 single protostars, 9 close binaries (defined as pairs with projected separations <500 au), and 5 wide multiples, one of which is a triple system with two widely separated companions. The smallest separation we resolve is $0\farcs21$ (27.6 au), while the widest system in our observed sample spans $61''$, corresponding to a projected separation of $\sim$8772 au. When including systems from the extended sample, we identify an even wider multiple system with five components and projected separations up to 9138 au.

Table 5. Companion Separations

Source Pair	Separation (arcsec)	Separation (au)	Classes
04295+2251 A + 04295+2251 B*	0.116±0.010	18.622±1.578	(CI+CI)
04158+2805 A + 04158+2805 B	0.212±0.009	27.567±1.181	(CI+CI)
04239+2436 A + 04239+2436 B	0.240±0.001	30.745±0.131	(CI+CI)
04248+2612 A + 04248+2612 B	0.297±0.004	38.010±0.536	(CI+CI)
04264+2433 A + 04264+2433 B	0.394±0.005	50.489±0.658	(CI+CI)
04287+1801 A + 04287+1801 B	0.422±0.003	60.829±0.485	(CI+CI)
L1551NE A + L1551NE B	0.549±0.006	79.106±0.839	(CI+CI)
04381+2540 A + 04381+2540 B	0.609±0.001	85.200±0.195	(CI+CI)
Haro 6-28 A* + Haro 6-28 B*	0.663	107.728	(CII+CII)
04361+2547 A + 04361+2547 B*	0.228±0.001	31.975±0.190	(CI+CI)
04263+2426 A + 04263+2426 B	1.363±0.002	174.461±0.213	(CI+CI)
04489+3042 A + 04489+3042 B	2.994±0.212	467.139±32.997	(CI+CI)
04191+1523 A + 04191+1523 B	6.055±0.005	871.946±0.663	(CI+CI)
04325+2402 A + 04325+2402 B	8.116±0.004	1038.823±0.452	(CI+CI)
HP Tau G3* + HP Tau G2*	9.358	1647.022	(CII+CII)
IRAM 04191 A + IRAM 04191 B*	12.708±0.009	1829.900±1.320	(C0+C0)
04248+2612 C +(04248+2612 B + 04248+2612 A)	12.869±0.012	1647.211±1.538	[CI+(CI+CI)]
FS Tau A* + FS Tau B*	16.592	2322.900	(FS+CI)
04108+2803 A* + 04108+2803 B	19.874±0.001	2583.610±0.154	(FS+CI)
(HP Tau G3 + HP Tau G2)* + HP Tau*	17.111	3011.575	[(CII+CII)+FS]
XZ Tau* + HL Tau*	22.397	3299.022	(CII+CI)
04181+2654 A + 04181+2654 B*	31.897±0.002	5039.699±0.285	(CI+CI)
LkHα358* + (XZ Tau + HL Tau)*	47.042	6774.019	[FS+(CII+CI)]
DG Tau A + DG Tau B	53.815±0.004	6888.373±0.499	(FS+CI)
(04191+1523 A + 04191+1523 B)+(IRAM 04191 A + IRAM 04191 B*)	54.774±0.003	7887.484±0.378	[(CI+CI)+(C0+C0)]
[(HP Tau G3 + HP Tau G2)+ HP Tau]* +(Haro 6-28 B + Haro 6-28 A)*	51.922	9138.274	[[(CII+CII)+FS]+(CII+CII)]

Note—An asterisk (*) denotes a source from the extended sample. Separation entries without associated errors are from the added Taurus+ sources, where right ascension and declination values were retrieved from the literature or SIMBAD where positional uncertainties were not provided.

For some targets, detections were made at only one wavelength. IRAS 04181+2654 B was not detected with either ALMA or the VLA, while HH 30, IRAS 04489+3042, and IRAS 04248+2612 ABC were undetected with the VLA. We retain IRAS 04181+2654 B in the extended sample given its confirmation in previous high-resolution studies (M. S. Connelley et al. 2008). In all cases of non-detection, we proceeded with analysis based solely on the available ALMA or VLA data in the observed sample. Some systems also showed differing multiplicity signatures between instruments. For example, IRAS 04158+2805 and IRAS 04191+1523 appeared as single sources with the VLA but were resolved as binaries with ALMA. Conversely, DG Tau A and B were both detected with the VLA, but only DG Tau B was detected with ALMA due to ALMA's smaller field of view.

We detect dust continuum emission with disk-like morphology around nine sources in our sample. Six of these are consistent with circumstellar disks surrounding single protostars, three of which appear edge-on. The remaining three sources, 04158+2805, L1551 NE, and IRAS 04287+1801, exhibit circumbinary disks encompassing close binaries, each showing evidence of a large central cavity, supporting the idea that large cavities in the protostellar phase can be indicators of binary formation (P. D. Sheehan et al. 2020). Our ALMA observations of IRAS 04295+2610 show a disk with a large central cavity but no detected inner source, whereas our VLA data reveal a single central point source. Based on the disk morphology, we infer the presence of a close companion and examine archival higher-resolution data toward this target to test this possibility.

3.1. *Archival data results*

The image from the combined VLA B and A-configuration of IRAS 04295+2251 (Figure 6) shows a prominent point source and a faint extension to the west, this extension could be a secondary source that is becoming marginally-

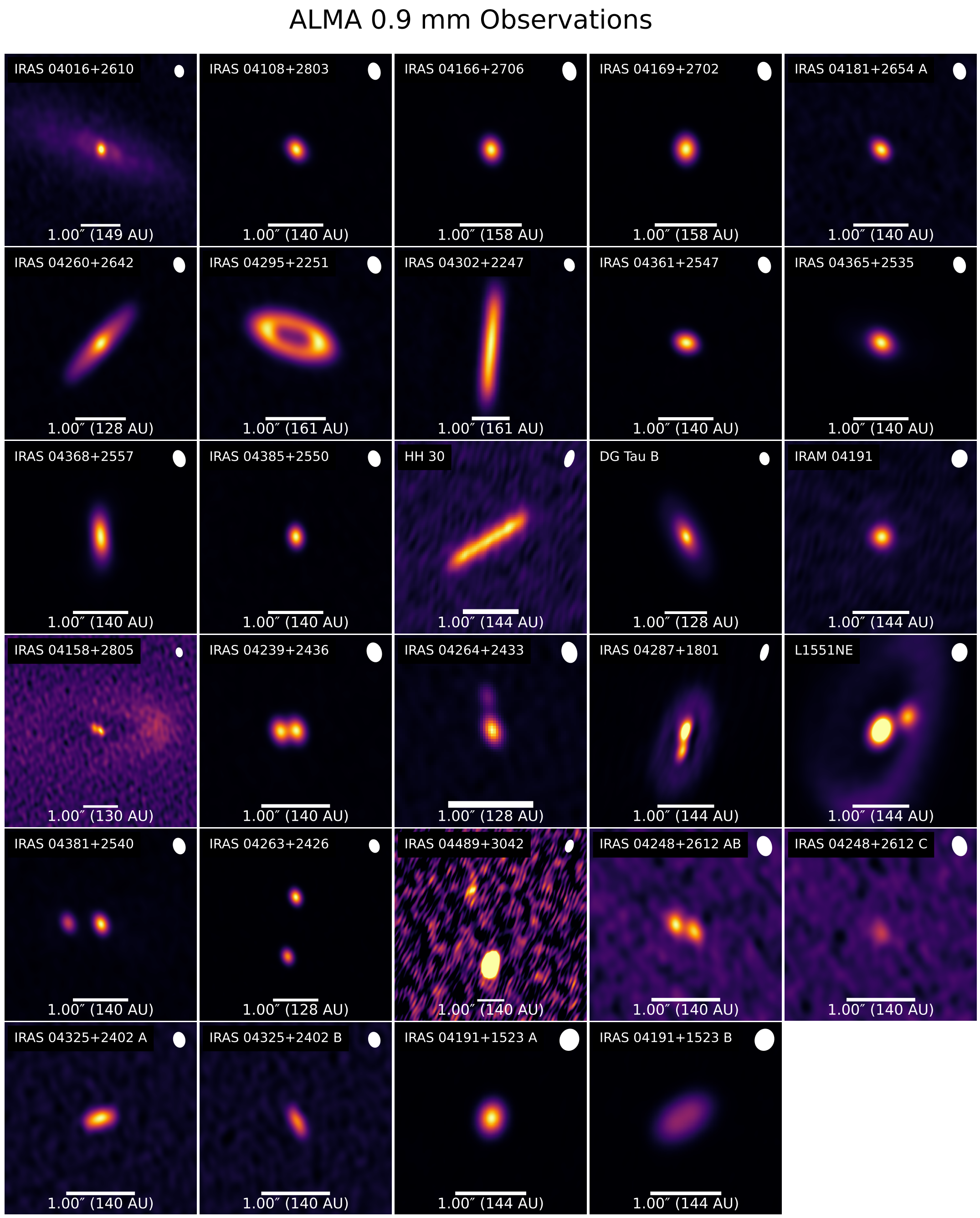

Figure 2. ALMA components with 0.9 mm continuum images of protostars in Taurus, displayed in order of single protostars, close binary systems, and zoomed-in views of wide multiples. Large scale views of these wide multiples can be found in Figure 4. Synthesized beams in the top-right of each panel give the angular resolution, and $1''$ scale bars are indicated.

resolved with the higher resolution. The proper motion correction of the source position to match the 2022 July observation puts the source within the observed central cavity of the disk detected at 0.9 mm.

We performed the following to assess the robustness of this extension being a secondary source within the central cavity of the disk around IRAS 04295+2251. We first fitted single Gaussian, with a size and orientation identical to the synthesized beam, to the main source. Subtracting this fit leaves a residual that has a peak of 61.2 μJy to the west, and this source has a significance of 5.8σ, relative to the noise level of 10.5 μJy. When fitted together as two Gaussians, the primary has a peak of 477μJy and the secondary has a peak of 75.8 μJy, with uncertainties on these

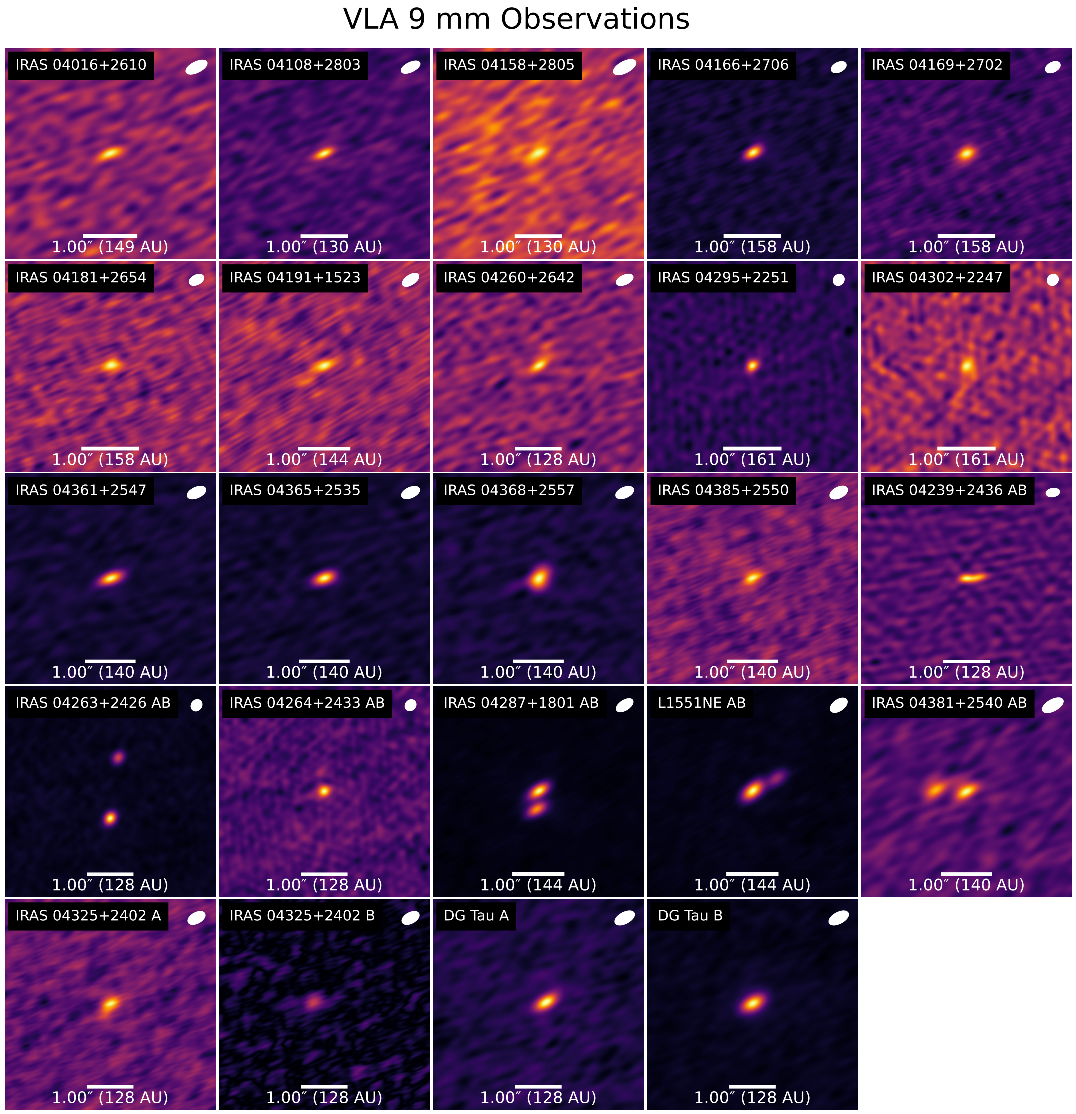

Figure 3. VLA components with 9 mm continuum images of protostars in Taurus, displayed in order of single protostars, close binary systems, and zoomed-in views of wide multiples. Large scale views of these wide multiples can be found in Figure 4. Synthesized beams in the top-right of each panel give the angular resolution, and $1''$ scale bars are indicated.

peaks of 12.8 μJy; thus the significance of the secondary peak when fitted is 5.9σ. We note that there are other noise peaks near the source that have peaks of 40.9 μJy beam^{-1} (to the west) and 36.3 μJy beam^{-1} (to the southeast), but both of these are below 4σ significance. Thus, while this secondary component might have a significance above 5σ, because of its blending and the fact that it stands out most after subtracting the main source, we only regard this as a potential companion, and include it in the Taurus+ sample. However, additional observations with longer integration time, better uv-coverage, and higher resolution will be able to confirm (or reject) the presence of this putative companion.

The presence of a companion source would offer a natural explanation for the presence of such a large cavity in the disk of IRAS 04295+2251, similar to the large cavity in the disk surrounding the binary source IRAS 04158+2805 and more widely separated sources like L1551NE and IRAS 04287+1801 (L1551 IRS5) that have rings surrounding their companions. The lack of detection by ALMA in the current observations may be due to blending with the surrounding disk and the fact that the circumstellar disks around the protostars may be very small limiting their brightnesses.

In the archival ALMA Band 6 data, we identify a secondary continuum source, hereafter IRAM 04191 B, located $\sim$12$.''$7 ($\sim$1800 au) northeast of IRAM 04191 A. This source has not been reported in previous studies of the system

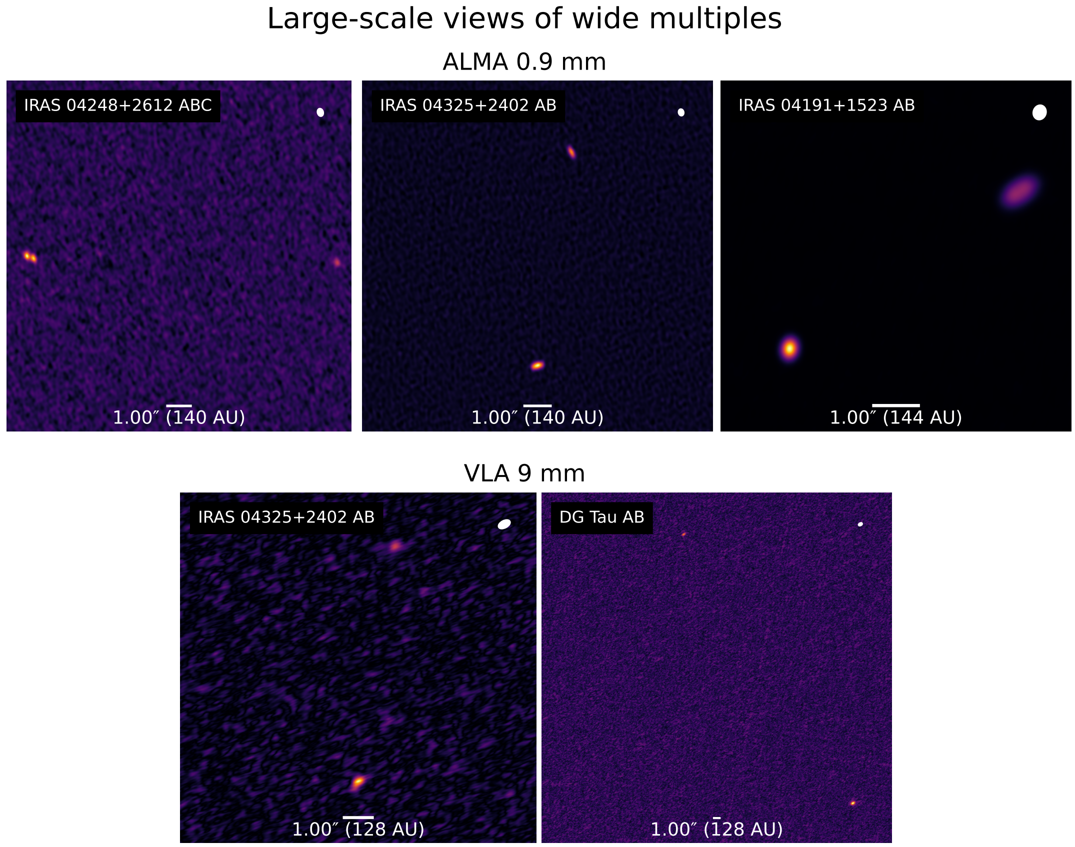

Figure 4. Large-scale views of the wide multiples from the ALMA observations (top) and VLA observations (bottom) to show their full projected separations. Small scale views of these wide multiples can be found in Figures 2 and 3. Synthesized beams in the top-right of each panel give the angular resolution, and $1''$ scale bars are indicated.

(e.g., P. André et al. 1999; X. Chen et al. 2012; M.-R. Kim et al. 2016; G. Kim et al. 2019; K. L. Luhman et al. 2010). IRAM 04191 B is detected at a peak intensity of $\sim$3.18 mJy beam^{-1} with an rms noise level of $\sim$0.22 mJy beam^{-1}. Because no infrared counterpart has been identified for this source, we adopt the Class 0 designation of IRAM 04191 A. As in the Band 7 ALMA observations (Figure 2), IRAM 04191 A appears as a compact point source. In the zoomed-in view of IRAM 04191 B, we detect faint extended emission that may trace an edge-on disk around this protostar.

4. MULTIPLICITY CHARACTERIZATION

4.1. *Bolometric Luminosities and Temperatures*

Figure 7 shows $L_{\rm bol}$ and $T_{\rm bol}$ of single and multiple protostar systems in Taurus. This plot highlights multiple systems with separations ranging from 18 to 10,000 au. For close multiples, where the components likely share a SED and thus identical $L_{\rm bol}$ and $T_{\rm bol}$ values, only one point is plotted per system. In contrast, wider multiples and higher-order systems are plotted with one point per component when the sources have distinct infrared fluxes.

Histograms along the top and right axes of the scatter plot show the distributions of $T_{\rm bol}$ and $L_{\rm bol}$, respectively. In the original sample, single protostars had a median luminosity of 0.68 $L_\odot$, while multiples had a median of 0.63 $L_\odot$. In the Taurus+ sample, the medians were 0.43 $L_\odot$ for singles and 0.85 $L_\odot$ for multiples. Thus, for the extended sample, we find that the median luminosity of multiple systems is approximately twice that of single systems. However, the shapes of the distributions are quite course due to the small sample size. The $T_{\rm bol}$ distributions appear less correlated, with no clear similarities in shape between singles and multiples. These distributions show that our sample is dominated by Class I protostars and not particularly biased toward very young or very evolved protostars.

To assess the significance of these differences, we performed a KS test comparing the $L_{\rm bol}$ distributions of singles and multiples. We could not reject the null hypothesis that the bolometric luminosity distributions of single and multiple

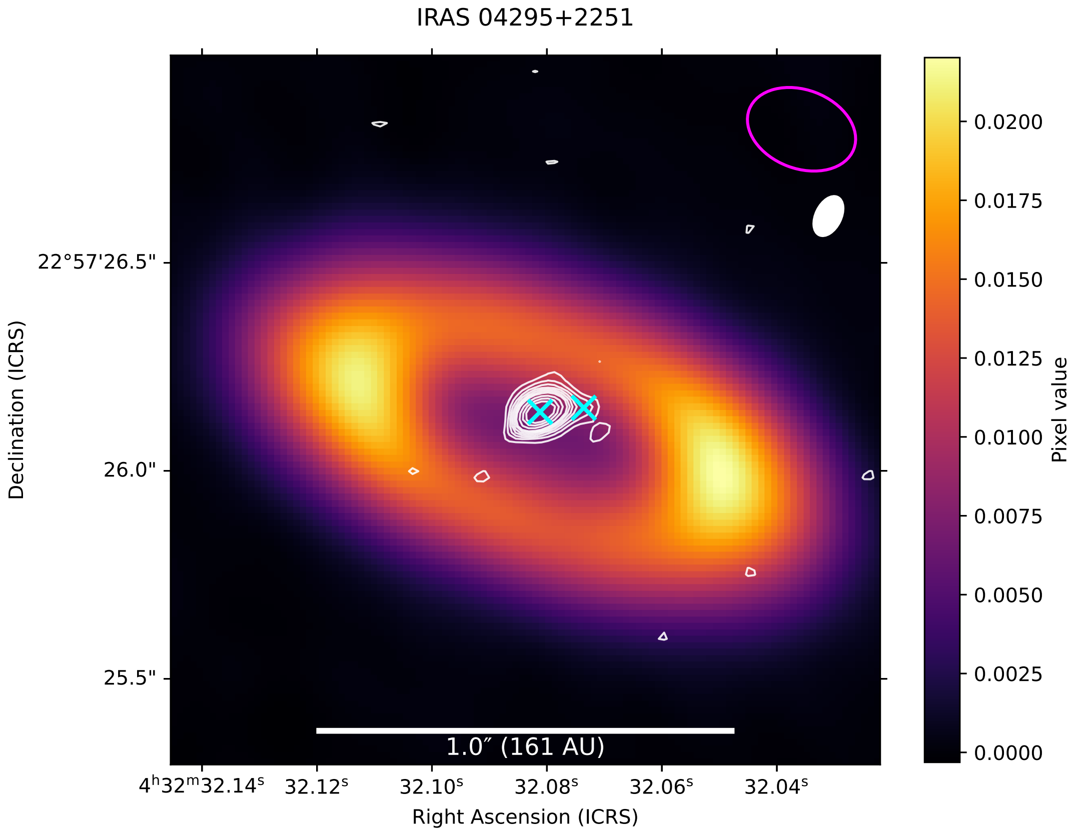

Figure 5. The magenta beam in the upper-right corner represents the synthesized beam of the ALMA 0.9 mm continuum image, which traces the disk with a large cavity. The filled white beam just below corresponds to the archival VLA 9 mm data used for the overlaid contours, drawn at multiples of the map rms ($\sigma = 10.5$ μJy). The cyan crosses mark the positions of the two proposed protostars, separated by 18 au.

systems are drawn from the same parent population for either sample, with $p = 0.90$ for the original sample and $p = 0.061$ for the extended sample.

In comparison, Orion and Perseus show stronger trends. In both regions, median luminosities of multiple systems are roughly three times higher than those of single systems ($\sim$3 $L_\odot$ vs. $\sim$1 $L_\odot$), and the KS test for Orion yielded a p-value < 0.01, suggesting statistically significant differences. For Perseus, the result is more marginal, with a p-value of 0.025, just above the threshold for significance. The $T_{\rm bol}$ distributions for single and multiple protostars are statistically consistent with being drawn from the same parent population.

The expectation that multiple protostars should be more luminous arises in part because accretion onto several components releases more total energy than accretion onto a single star at the same rate. Multiplicity may also correlate with initial core mass, since more massive cores are more prone to fragmentation and capable of forming brighter systems (G. Duchêne & A. Kraus 2013). In addition, higher-mass protostars themselves are more likely to be in multiple systems, which could further augment the luminosities observed in multiples. The link between stellar mass and multiplicity is well documented (G. Duchêne & A. Kraus 2013). While some of these higher-mass main-sequence stars may have acquired companions via capture, H. Beuther et al. (2025) describe how disks around massive stars are more prone to instability and fragmentation, contributing to their higher multiplicity rates. In Taurus, we observe only a slight skew toward higher luminosities in multiples, and that trend is not statistically significant in the extended sample and not observed at all in the original sample.

4.2. *Multiplicity Statistics*

Table 6 presents the multiplicity statistics for Taurus, including both the original and extended samples, and compares them to the star-forming regions of Orion and Perseus. Across the full separation range of 18–10,000 au, we find an MF of 0.50 and a CF of 0.58 in our observed Taurus sample. These values are notably higher than those reported

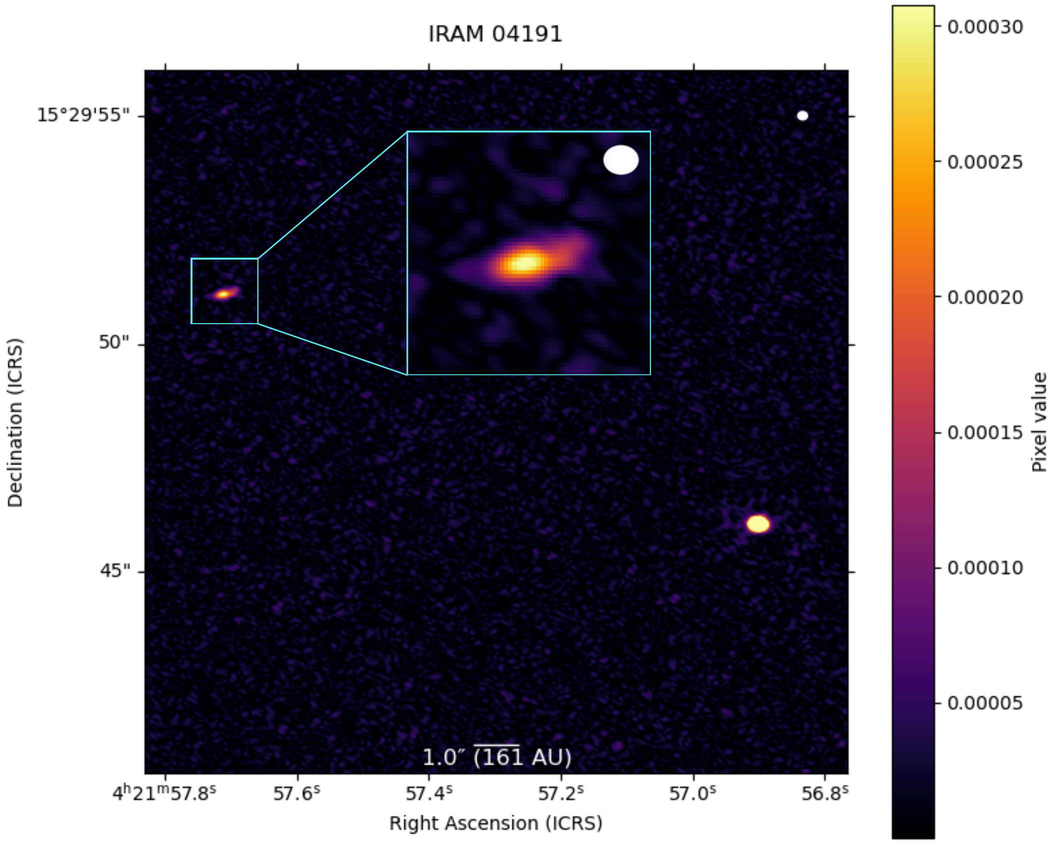

Figure 6. This ALMA 1.3 mm continuum image has a synthesized beam corresponding to a resolution of 0.″154. The source on the lower right is IRAM 04191 A which was detected in our original Band 7 observations (Figure 2), while the source in the upper left is a new detection, not previously reported, which we designate as IRAM 04191 B. A zoomed-in view of this new source is shown in the cyan box.

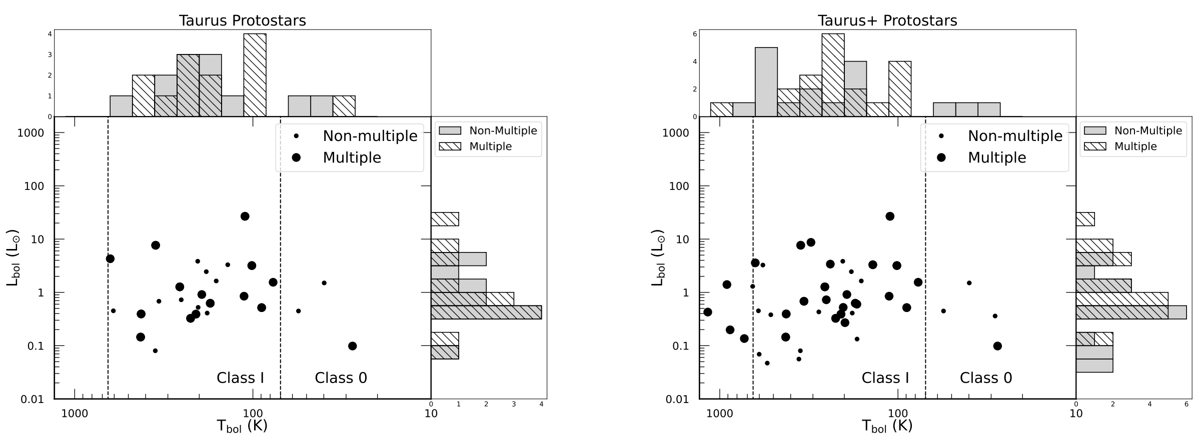

Figure 7. Bolometric luminosity ($L_{\rm bol}$) versus bolometric temperature ($T_{\rm bol}$) for protostars in Taurus. The left panel shows the original sample, and the right panel includes sources from the extended sample. In both plots, small points represent single protostar systems, while larger points indicate components of multiple systems. Histograms along the top and right axes show the distributions of $T_{\rm bol}$ and $L_{\rm bol}$ for singles (gray fill) and multiples (hatched). Vertical dashed lines mark approximate boundaries between Class 0 and Class I ($T_{\rm bol}$= 70 K) and between Class I and Class II ($T_{\rm bol}$ = 650 K). Flat Spectrum protostars can appear throughout the Class I range and sometimes into Class II as that classification is based on the spectral index from 2-24 μm and not $T_{\rm bol}$.

Table 6. Multiplicity Statistics

Region	Separation Range (au)	Counts	MF	CSF
Taurus All	18 - 10000	12:10:2:0:0:0:0:0:0:0	$0.50^{+0.12}_{-0.12}$	$0.58^{+0.20}_{-0.20}$
Taurus+ All	18 - 10000	17:15:2:1:1:0:0:0:0:0	$0.53^{+0.09}_{-0.10}$	$0.72^{+0.19}_{-0.19}$
Orion All	18 - 10000	217:69:11:2:4:1:0:0:0:1	$0.29^{+0.03}_{-0.03}$	$0.42^{+0.03}_{-0.03}$
Perseus All	18 - 10000	45:18:4:2:1:0:0:0:0:0	$0.36^{+0.07}_{-0.06}$	$0.51^{+0.11}_{-0.11}$
Taurus All	18 - 1000	18:10:0:0:0:0:0:0:0:0	$0.36^{+0.11}_{-0.10}$	$0.36^{+0.11}_{-0.10}$
Taurus+ All	18 - 1000	36:13:0:0:0:0:0:0:0:0	$0.27^{+0.08}_{-0.07}$	$0.27^{+0.08}_{-0.07}$
Orion All	18 - 1000	306:57:4:0:0:0:0:0:0:0	$0.17^{+0.02}_{-0.02}$	$0.18^{+0.02}_{-0.02}$
Perseus All	18 - 1000	63:20:1:0:0:0:0:0:0:0	$0.25^{+0.06}_{-0.05}$	$0.26^{+0.06}_{-0.05}$
Taurus All	18 - 500	20:9:0:0:0:0:0:0:0:0	$0.31^{+0.11}_{-0.09}$	$0.31^{+0.11}_{-0.09}$
Taurus+ All	18 - 500	38:12:0:0:0:0:0:0:0:0	$0.24^{+0.08}_{-0.06}$	$0.24^{+0.08}_{-0.06}$
Orion All	18 - 500	330:48:2:0:0:0:0:0:0:0	$0.13^{+0.02}_{-0.02}$	$0.14^{+0.02}_{-0.02}$
Perseus All	18 - 500	71:16:1:0:0:0:0:0:0:0	$0.19^{+0.05}_{-0.04}$	$0.20^{+0.05}_{-0.04}$
Taurus All	100 - 1000	25:3:0:0:0:0:0:0:0:0	$0.09^{+0.08}_{-0.05}$	$0.09^{+0.08}_{-0.05}$
Taurus+ All	100 - 1000	45:4:0:0:0:0:0:0:0:0	$0.07^{+0.05}_{-0.03}$	$0.07^{+0.05}_{-0.03}$
Orion All	100 - 1000	327:37:3:0:0:0:0:0:0:0	$0.10^{+0.02}_{-0.02}$	$0.11^{+0.02}_{-0.02}$
Perseus All	100 - 1000	72:11:1:0:0:0:0:0:0:0	$0.13^{+0.04}_{-0.04}$	$0.14^{+0.05}_{-0.04}$

for Orion (MF = 0.29, CF = 0.42) and Perseus (MF = 0.36, CF = 0.51), indicating that a larger fraction of systems in Taurus are multiples rather than singles. Although the differences are only significant at the $\sim$1–2σ level, this trend holds for the extended sample as well, with Taurus+ showing MF = 0.53 and CF = 0.72.

However, when we restrict the sample to farther separations (100-10000 au) the MF and CF in Taurus and Taurus+ decrease substantially and become comparable to those in Orion and Perseus. This suggests that Taurus does not have an excess of wide multiples but instead exhibits more closely separated systems.

The CF in Taurus closely tracks the MF across all separation ranges, implying that most systems are simple binaries rather than higher-order multiples. An exception to this trend appears in the extended sample at wide separations, where there are several higher order systems comprised of 4 or 5 components.

4.3. *Separation Distributions*

We also characterize the typical separations between protostellar companions and explore how these distributions may reflect underlying star formation processes. To do this, we analyze the full set of projected separations measured for multiple systems (see Table 5), using the methodology outlined in Section 2.8. We visualize the separation distributions using both histograms and cumulative distribution functions (CDFs) to investigate more subtle differences between populations. Histograms are binned uniformly in log space, with a width of 0.25 dex in $\log_{10}$(separation in au). While in principle the distributions could be broken down by evolutionary class, we include all Class 0, I, Flat Spectrum, and II sources together for this analysis because of the limited sample size in Taurus.

4.3.1. *Taurus Samples*

We first compare the separation distributions of the observed Taurus sample and the Taurus+ sample, as shown in Figure 8. In the observed sample, the distribution peaks at $\sim$75 au. There is a noticeable drop in the number of systems around 200 au, and a broader gap in the distribution around 3000 au.

In the Taurus+ sample, the $\sim$75 au peak persists, though it appears slightly less pronounced. The gap near 200 au also remains, and there are fewer multiples between 200-1000 au. However, the extended sample reveals additional structure, with rising multiplicity toward 10,000 au. This rise at the widest separations is likely driven primarily by the original sample, but the inclusion of additional sources in the extended sample helps fill in the gap around 3000 au. The result is a broader and more continuous distribution, suggesting a population of wide companions, similar in number to the <300 au multiples.

The cumulative distribution functions in Figure 8 highlight these differences. The observed Taurus sample has a steeper rise at separations <300 au and then becomes more shallow at wider separations. In other words, the observed sample is more heavily weighted toward close multiples. The flattening of both CDFs near 200–300 au reflects the lack of systems at intermediate separations, while steeper slopes below and above that range indicate some clustering at small and wide separations, respectively. These trends are similar in the Taurus+ sample, but the CDF only reaches 0.5 at 1000 au reflecting the greater contribution to wide multiples in the Taurus+ sample.

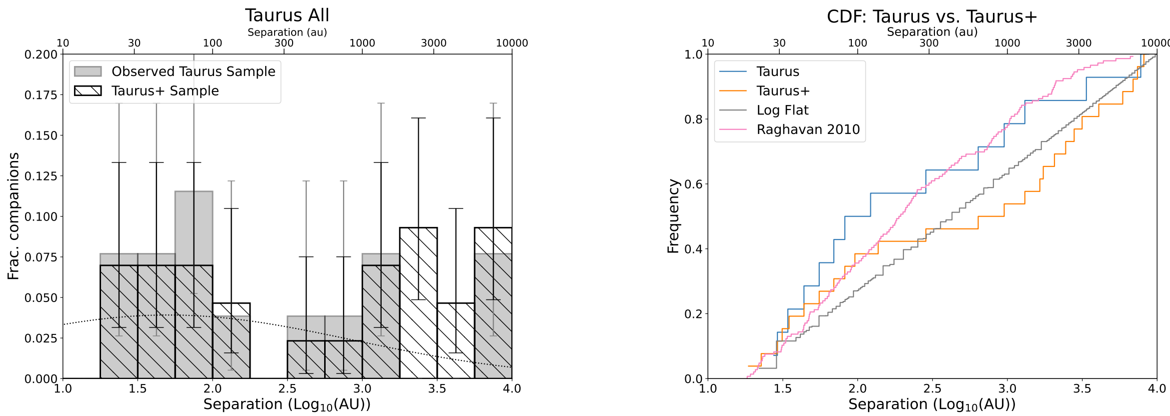

Figure 8. The panel on the left shows histograms of companion separation distributions for the observed Taurus sample (gray) and the extended Taurus+ sample (hatched). The Gaussian fits of the separation distribution of solar-type main sequence field stars from D. Raghavan et al. (2010) is also included as a dotted line for comparison. Each bin's error bar is calculated based on binomial statistics, following the method outlined in Section 2.8.2. The right panel displays cumulative distribution functions of separations, plotted in terms of companion frequency, for Class 0/I/FS protostars in the Taurus and Taurus+ samples. For reference, a log-flat distribution is shown in gray, and the field star distribution from D. Raghavan et al. (2010) is overlaid in red.

4.3.2. *Orion & Perseus Comparisons*

We compare the Taurus separation distributions to those of Orion and Perseus in Figure 9 (histograms and CDFs). Both Orion and Perseus were corrected for YSO density, and the multiple systems were weighted by the probability that each companion was real. This correction primarily affected systems at large separations but had little impact on close companions. For comparison, we adopted the conservative probabilities.

Both Orion and Perseus exhibit a characteristic double-peaked structure in their histograms (top panel, Figure 9), with a peak near 75 au and a second broader rise near 4000 au. These peaks are separated by a local minimum around 300 au, giving the appearance of a bimodal separation distribution. This bimodality is more pronounced in Orion, but still apparent in Perseus. In contrast, the original Taurus sample is dominated by a single peak at close separations ($\sim$75 au) and sparsely populates the wide-separations, making it difficult to identify any significant structure beyond a few hundred au. However, the extended Taurus+ sample fills in the gap near 3000 au, potentially revealing a broader bimodal distribution more reminiscent of Orion and Perseus.

J. J. Tobin et al. (2022) showed that in Orion and Perseus, this bimodality is largely driven by Class 0 systems, whereas more evolved Class I and Flat Spectrum sources exhibit flatter separation distributions. Taurus, which includes only five Class 0 sources (just two of which comprise a single binary), may therefore lack the strong bimodal signal seen in Orion.

The CDFs displayed in the bottom panel of Figure 9 offer additional insight by showing the companion frequency per system as a function of separation. At separations $\leq$ 300au, the original Taurus sample has the steepest rise—about 60% of its companions lie within this range—indicating a strong concentration of close binaries. Taurus+ and Perseus both reach $\sim$40% by 300 au, while Orion remains the shallowest, with only $\sim$30% of companions at close separations. Notably, Orion is the only region in which wide companions outnumber close ones, explaining why its CDF remains lowest across the entire separation range.

4.3.3. *Comparison with older Taurus Population*

We compare our Taurus separation distributions to the more evolved Class II and III stars in Taurus from A. L. Kraus et al. (2011), shown in Figure 10. The histogram displays the full sample, which includes stars across a range of masses (0.25-2.5 $M_\odot$), along with separate histograms for the low-mass ($<$0.7 $M_\odot$) and high-mass ($>$0.7 $M_\odot$) primary

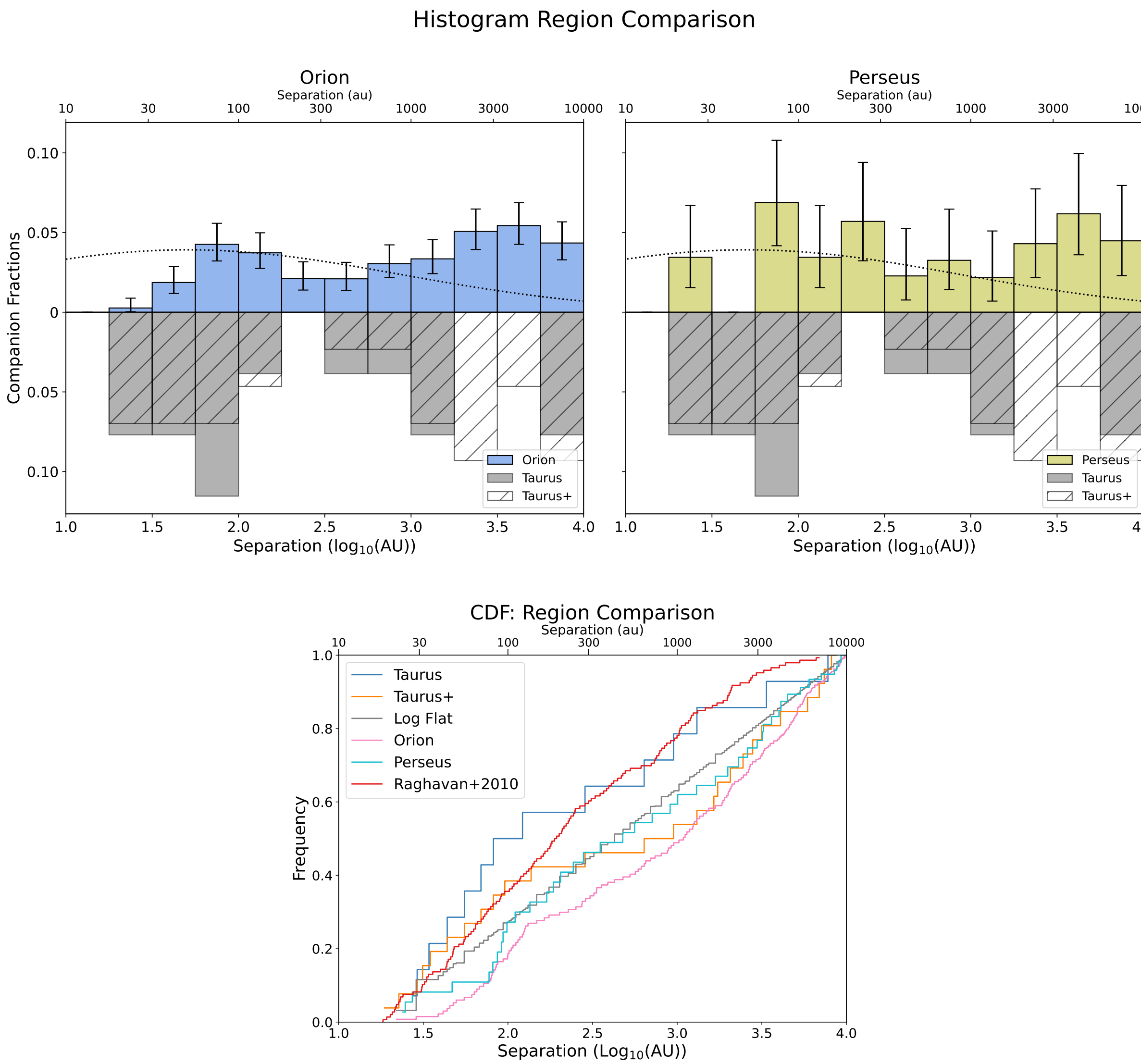

Figure 9. Mirrored histograms (top panels) show the companion separation distributions for Orion (left, blue) and Perseus (right, green). The corresponding Taurus and Taurus+ distributions are mirrored below each. For Orion and Perseus, we adopt the contamination-corrected samples. The dotted line indicates the distribution of solar-type main-sequence field stars from D. Raghavan et al. (2010). The bottom panel presents the CDFs of companion separations, expressed in terms of companion frequency, for protostars in Taurus, Orion, and Perseus. The red curve represents the CDF for solar-type field stars from D. Raghavan et al. (2010), while the gray curve shows a reference log-flat distribution.

subsamples. The full A. L. Kraus et al. (2011) sample displays a peak around 40 au, visible in both the low-mass and high-mass subsamples. At wider separations, the distribution flattens out, balancing the lack of wide companions in the low-mass sample with an excess in the high-mass sample. The low-mass systems are strongly concentrated at close separations and show a scarcity of wide multiples. Due to resolution limits, we cannot assess the innermost separation bins (3-18 au) probed by A. L. Kraus et al. (2011).

The Taurus+ sample resembles the A. L. Kraus et al. (2011) high-mass subsample more closely than the low-mass one, with both distributions showing a lack of intermediate separation companions and possible peaks at both close and wide separations. However, the lull in intermediate Taurus+ companions occurs at slightly wider separations (∼250 au) compared to the ∼75 au turnover in the A. L. Kraus et al. (2011) high-mass sample. These similarities are also apparent in the CDFs shown in Figure 11, where both Taurus+ and A. L. Kraus et al. (2011) high-mass share a similar overall shape and curvature. The A. L. Kraus et al. (2011) low-mass sample, however, has the steepest CDF rise and the shortest overall separation range, further reinforcing the dominance of close companions. Since we cannot compare the closest separations probed by A. L. Kraus et al. (2011), we omit companions within ∼30 au from the CDF.

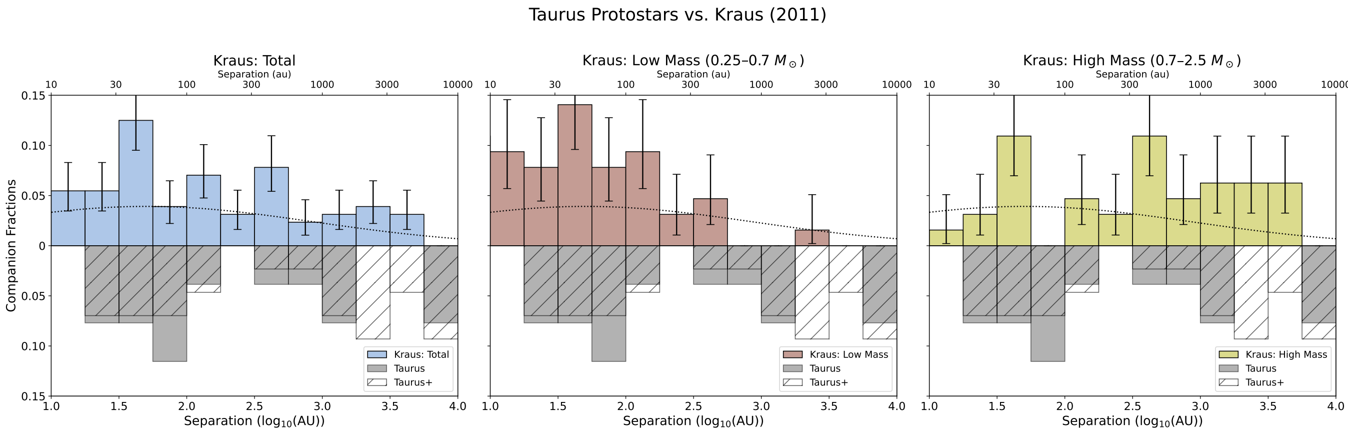

Figure 10. Mirrored histograms showing separation distributions for the 3 samples of more evolved protostars from A. L. Kraus et al. (2011): Total, Low Mass, High Mass (left to right). Mirrored below each A. L. Kraus et al. (2011) sample is the Taurus & Taurus+ histograms. Since the total number of systems used to calculate the CF for the low- and high-mass samples was not specified in A. L. Kraus et al. (2011), we assumed 64—half of the full sample—based on the roughly equal numbers of low- and high-mass systems.

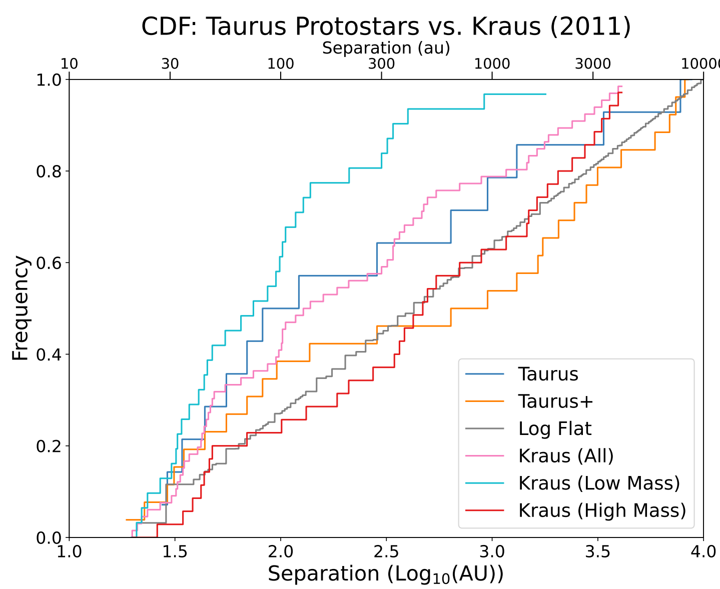

Figure 11. Cumulative distribution functions of companion separations in Taurus. The dark blue and orange curves show Class 0/I protostars from our Taurus and Taurus+ samples, respectively. The more evolved Class II/III systems from A. L. Kraus et al. (2011) are also shown: the full Kraus sample in pink, low-mass primaries (0.25–0.7 $M_\odot$) in light blue, and high-mass primaries (0.7–2.5 $M_\odot$) in red. The gray curve represents a reference log-flat distribution.

4.4. *Quantitative Separation Distribution Comparisons*

To assess whether different protostellar separation distributions are statistically consistent with one another, we performed KS tests on the CDFs of companion separations and Anderson-Darling tests on the separation distributions between 18–10,000 au. The results are presented in Table 7. We adopt a significance threshold of $p<0.05$ to indicate that

Table 7. Separation Distributions Statistical Comparisons

Comparison	Samples	KS Test p-value	AD Test p-value
Taurus vs. Orion	14, 128	0.057	0.012
Taurus vs. Perseus	14, 38	0.143	0.087
Taurus vs. Log-flat	14	0.290	0.136
Taurus vs. Raghavan+2010	14, 135	0.464	0.427
Taurus vs. Kraus+2011	14, 62	0.879	0.650
Taurus vs. Kraus+2011 LM	14, 28	0.173	0.087
Taurus vs. Kraus+2011 HM	14, 34	0.294	0.390
Taurus+ vs. Orion	26, 128	0.261	0.090
Taurus+ vs. Perseus	26, 38	0.478	0.464
Taurus+ vs. Log-flat	26	0.176	0.403
Taurus+ vs. Raghavan+2010	26, 135	0.014	0.006
Taurus+ vs. Kraus+2011	26, 62	0.028	0.016
Taurus+ vs. Kraus+2011 LM	26, 28	0.001	0.001
Taurus+ vs. Kraus+2011 HM	26, 34	0.486	0.213

Note—Results of the Kolmogorov–Smirnov and Anderson–Darling tests were applied to distributions encompassing all classes over the separation range 18–10,000 au.

two samples are unlikely to be drawn from the same parent distribution. Among the comparisons tested, most p-values exceed this threshold, suggesting broad consistency between distributions. However, three cases show statistically significant differences.

First, the Taurus+ sample differs significantly from the field solar-type binary population of D. Raghavan et al. (2010), with a KS p-value of 0.014 and even more significant AD p-value of 0.006, allowing us to rule out the null hypothesis. This likely reflects the excess of wide multiples in Taurus+ compared to the narrower separation range characteristic of field multiples. Similarly, the Taurus+ sample differs from both the full A. L. Kraus et al. (2011) sample of Class II/III stars ($p_{\rm KS} = 0.028$, $p_{\rm AD} = 0.016$) and its low-mass subsample ($p_{\rm KS} = 0.001$, $p_{\rm AD} = 0.001$). These differences are again likely driven by the relative abundance of wide systems in Taurus+ and the lack of wide companions among more evolved, low-mass stars in the A. L. Kraus et al. (2011) sample. The Taurus sample is unlikely to be drawn from the same parent distribution as Orion, based on the AD test (p = 0.012), whereas this difference is not observed in the Taurus+ vs. Orion comparison, but still has low probability.

In contrast, there is no evidence that the Taurus or Taurus+ distributions differ significantly from those Perseus or the high-mass component of the A. L. Kraus et al. (2011) sample. For example, the KS and AD p-values for Taurus+ vs. A. L. Kraus et al. (2011) high-mass are 0.486 and 0.213, respectively, indicating consistency. The observed Taurus sample also matches well with the full A. L. Kraus et al. (2011) sample ($p_{\rm KS} = 0.879$, $p_{\rm AD} = 0.650$), though the small sample size (N = 14) limits the statistical power of this comparison.

We also tested both Taurus samples against a log-flat separation distribution, which is often referred to as Öpik's law (E. Öpik 1924), and has been shown to describe companion separation populations in some regions (e.g., M. B. N. Kouwenhoven et al. (2007)). In both cases, the results were inconclusive ($p > 0.05$), and we cannot reject the hypothesis that the data are consistent with a log-flat distribution. This suggests that while hints of bimodality are present in the histograms, they are not statistically significant enough to definitively rule out a flat distribution in log space.

Among all comparisons, the null hypothesis can only be rejected ($p < 0.01$) for the Taurus+ vs. A. L. Kraus et al. (2011) low-mass samples, and the Taurus+ vs. Raghavan 2010 samples, suggesting they are not drawn from the same parent distributions.

5. DISCUSSION

We will discuss what the Taurus multiplicity measurements imply about how companions form and evolve. Taurus provides a useful contrast to denser regions: its low stellar density and nearly complete protostellar census help isolate environmental effects. Our main tool is the distribution of companion separations, which we use as a proxy for formation pathway (core vs. disk fragmentation) and early orbital evolution (e.g., migration). We compare the Taurus and Taurus+ samples to a simple log-flat reference in separation and to results from Orion and Perseus, and we relate them to more evolved Taurus and field populations to place the Taurus results in a broader context.

5.1. *Implications of the Observed Separation Distributions*

Multiple star formation is now thought to proceed through two dominant physical mechanisms, each operating at different spatial scales. Core fragmentation typically occurs on scales of thousands of au, resulting from instabilities within a collapsing molecular core (S. S. R. Offner et al. 2010; A. T. Lee et al. 2019). In contrast, disk fragmentation takes place within the rotationally supported circumstellar disk and is expected to form companions at separations of hundreds of au or less (K. M. Kratter et al. 2010). As a result, if both mechanisms are at play, we expect the resulting multiplicity to span from tens to thousands of au, which is consistent with the full range of separations observed in our Taurus samples.

While these mechanisms act during the formation stage, subsequent dynamical evolution can significantly reshape the observed distribution. Simulations by S. S. R. Offner et al. (2010), A. T. Lee et al. (2019) and R. L. Kuruwita & T. Haugbølle (2023) demonstrate that wide companions, initially formed at separations beyond ∼1000 au, can migrate inward to separations below 100 au on timescales of less than a few 100,000 years. These changes are often driven by gravitational interactions with surrounding gas or other protostellar components, and in higher-order systems, dynamical instabilities can lead to ejections that further alter system architectures (B. Reipurth & S. Mikkola 2012). Because such evolution happens rapidly, the study of young protostellar populations offers a unique opportunity to observe multiplicity properties closer to their primordial state, providing critical insight into the conditions and mechanisms that govern multiple star formation. Observational studies in other regions support the presence of these dual formation pathways. In Perseus, the survey by J. J. Tobin et al. (2016) revealed a bimodal separation distribution, with peaks near 75 au and 3000 au. They suggested that the wide and close peaks reflect the roles of core fragmentation and disk fragmentation, respectively. Additionally, J. J. Tobin et al. (2016, 2018) and N. K. Reynolds et al. (2021, 2024) found that many close multiples reside within rotating disk-like structures, and that at least one disk is likely currently gravitationally unstable. Moreover, disk alignments at close separations suggest that these systems may have formed through disk fragmentation.

The Orion survey by J. J. Tobin et al. (2022) strengthened this idea. By sampling a large population of Class 0, Class I, and Flat Spectrum sources, they found that the bimodal structure is primarily driven by the Class 0 systems, while more evolved sources displayed flatter or more uniform distributions. This suggests that the primordial multiplicity distribution may begin with distinct peaks associated with both fragmentation modes, which are then gradually smoothed out by migration and dynamical evolution. Supporting evidence for this bimodal morphology is also found in earlier work by M. S. Connelley et al. (2008), who identified a similar valley between close and wide companions in samples of Class I sources across multiple star-forming regions. Its repeated appearance in different star-forming regions suggests that two distinct mechanisms may be responsible for this recurrent feature, rather than it being a short-lived or transient pattern.

Figure 8 suggests that Taurus may also exhibit a bimodal separation distribution, with the observed sample showing a peak at close separations and a local minimum at intermediate scales, while the extended Taurus+ sample reveals an increase in multiplicity toward 2000–10,000 au. However, statistical tests indicate that we cannot definitively rule out a log-flat distribution for either the Taurus or Taurus+ samples.

Statistically, we find that we cannot rule out the null hypothesis that the separation distributions in Taurus or Taurus+ are consistent with those in Orion or Perseus. The observed sample yields $p_{\rm KS} = 0.057$ for Orion and $p_{\rm KS} = 0.143$ for Perseus, while the extended Taurus+ sample shows even less tension, with $p_{\rm KS} = 0.261$ (Orion) and $p_{\rm KS} = 0.478$ (Perseus). While these results do not allow us to claim a statistically significant similarity or difference, the qualitatively similar appearance of the distributions is compelling.

We suggest that the Taurus separation distribution may reflect a similar bimodal structure to that seen in Orion and Perseus. The primary limitation here is likely small number statistics: although our sample is nearly complete, the absolute number of protostellar systems in Taurus is much lower than in Orion or Perseus. With fewer sources, statistical fluctuations become more prominent, and clear bimodal features are more difficult to resolve.

Ideally, to recover the primordial distribution, one would analyze only Class 0 protostars. However, Taurus contains only five Class 0 systems, and just one of these is in a single binary system. As a result, the separation distribution in Taurus is predominantly shaped by Class I and Flat Spectrum sources. This contrasts with Orion, where the bimodality is driven by the Class 0 population, while Class I and Flat Spectrum sources show a more uniform distribution. The implication is that in Orion, dynamical evolution has smoothed out the bimodality in later stages—while in Taurus, the older protostars are less dynamically perturbed and may still show their primordial distribution with abundances of close and wide separation companions.

Environmental conditions likely play a role in this distinction. D. Guszejnov et al. (2023) investigated how different environments influence multiplicity, and found that stellar density is a particularly important factor. In high-density regions like Orion, dynamical interactions between young stars are frequent, leading to disrupted or reconfigured systems and increasing the likelihood of orbital migration that populates the close-binary regime. In contrast, Taurus is a low-density region, with fewer dynamical interactions. As a result, multiple systems may be more stable and retain their initial separations, potentially leading to wider multiples on average.

This environmental difference could explain why the bimodal distribution in Orion is driven by Class 0 sources. These systems are actively being reshaped by their environment, while the Class I and Flat Spectrum sources in Taurus retain a distribution that may still reflect their formation. Although, we do not observe a statistically significant excess of wide companions in Taurus relative to Orion or Perseus.

5.2. *Formation Mechanisms*

There is ongoing debate regarding how much disk fragmentation contributes to the formation of multiple star systems, particularly for low-mass stars. S. S. R. Offner et al. (2010) argues that turbulent core fragmentation is the dominant formation mechanism for low-mass multiples. Their radiation-hydrodynamic simulations show that while disks may fragment if mass is delivered faster than it can accrete onto the star (C. D. Matzner & Y. Levin 2005), radiative feedback raises disk temperatures and generally suppresses such fragmentation. They therefore claim that under realistic conditions, low-mass multiples arise primarily through core fragmentation of the parent core, consistent with the observed lower binary fractions among M dwarfs (C. J. Lada 2006).

Other simulations suggest that core fragmentation alone may not fully explain the observed multiplicity distributions, particularly the abundance of close multiples (<500 au). A. T. Lee et al. (2019) used gravo-magnetohydrodynamic simulations to model multiple star formation through turbulent core fragmentation, but lacked the resolution to capture disk fragmentation. Their results match the observed separation distributions between ∼100–3000 au but significantly underpredict companions at <100 au. They argue that without disk fragmentation, their models cannot reproduce the close-separation peak seen in observations such as J. J. Tobin et al. (2016, 2022), which report CFs 2–3× higher than those predicted by the simulations of A. T. Lee et al. (2019) for separations <500 au. This suggests that either disk fragmentation or post-formation orbital evolution is needed to account for the observed population of close multiples, a conclusion that applies to our Taurus sample as well.

R. L. Kuruwita & T. Haugbølle (2023) similarly simulate core fragmentation, but at a higher resolution that allows them to follow orbital evolution down to ∼20 au. However, disk fragmentation cannot be captured in their simulations due to limited resolution. Even so, their results show that dynamical processes, whether through interactions such as ejections and captures, or simply through the initial velocities and boundedness of neighboring fragments, can transform initially wide companions into close multiples, producing a bimodal separation distribution consistent with observations in regions like Perseus. They argue that massive disks, which are more susceptible to fragmentation, are rare (E. G. Cox et al. 2017; M. Ansdell et al. 2018), and instead emphasize the role of environmental density in reshaping wide systems formed through core fragmentation. In dense regions, frequent close encounters further drive orbital evolution and shrink binary separations, whereas in lower-density regions like Taurus, the scarcity of interactions, or perhaps simply fewer initial core fragmentation events, may limit the formation of close multiples. This reduced frequency of interactions could limit the efficiency of dynamical inspiral, raising doubts about whether the model from R. L. Kuruwita & T. Haugbølle (2023) could alone account for the observed number of close multiples in Taurus. Our results suggest that in less dense environments, additional processes such as disk fragmentation may still be necessary to explain the full multiplicity distribution.

5.3. *Elevated Multiplicity in Taurus*

A key distinction between Taurus and other star-forming regions is its elevated MF and CF. This rise is likely driven by environmental conditions, particularly stellar density. Simulations by D. Guszejnov et al. (2023) identified stellar density as the dominant environmental factor affecting multiplicity properties, showing a clear anti-correlation: lower-density environments produce higher MFs. This trend is consistent with our findings. In our observed Taurus sample, we find an MF of 0.50, compared to 0.36 in Perseus and 0.29 in Orion. The CF in Taurus is similarly elevated at 0.58, while Perseus and Orion show lower values of 0.51 and 0.42, respectively. These differences are even stronger in our Taurus+ sample: the MF rises to 0.53, which is 24% higher than Orion and 17% higher than Perseus, while the CF rises to 0.72, 30% higher than Orion and 21% higher than Perseus. These differences in MF and CF are only significant at the $\sim 1-2\sigma$ level, as the uncertainties in the Taurus samples are higher due to its smaller protostellar population. Although we consider the Taurus+ sample to be nearly complete, the results should be interpreted as suggestive trends, given the limitations of small-number statistics and the fact that some sources have not been observed at sub-arcsecond resolution.

These results are consistent with the overall trend of high multiplicity in Taurus. For example, the Class II and III multiplicity survey by A. L. Kraus et al. (2011) reported an MF of 0.73. However, approximately 10% of their multiples had separations below 18 au (below the resolution limits of this study). When we adjust their results to match our separation range, the total MF drops to 0.56, comparable to the 0.50 Taurus MF or the 0.53 Taurus+ MF. This suggests that elevated multiplicity is a persistent feature in this region, from the earliest stages through to more evolved systems.

This pattern aligns with expectations based on Taurus's low stellar density, estimated to be just 1–10 stars pc^{-3} (K. L. Luhman 2000). In contrast, the much denser environments of Orion and Perseus foster more frequent dynamical interactions, including close encounters, ejections, and companion exchanges. These processes tend to disrupt wide multiples and reduce the long-term multiplicity of systems, particularly for later-forming stars. While some stars in dense clusters may temporarily gain companions via capture, the overall effect of such dynamical processing is a suppression of stable multiplicity. In low-density environments like Taurus, however, systems may be more likely to retain their original companions, resulting in the higher observed MF and CF.

5.4. *Evolution*

While our study provides a snapshot of the multiplicity properties of young protostars in Taurus, it is important to recognize that these systems are unlikely to remain in their current configurations. Although the rate and degree of dynamical evolution may be slower or less extreme than in denser regions like Orion or Perseus, changes over time are still expected to reshape their separation distributions. A useful point of comparison is the more evolved stellar population in Taurus. A. L. Kraus et al. (2011) conducted a multiplicity survey of Class II and III stars in the region, dividing their sample into low-mass (0.25–0.7 $M_\odot$) and high-mass (0.7–2.5 $M_\odot$) subsamples. They reported an overall MF of 0.73, with values of 0.77 for the low-mass group and 0.62 for the high-mass group. After correcting these values to match the resolution limits of our survey (18-10000 au), we obtain an overall MF of 0.56, with the low- and high-mass subsamples yielding 0.59 and 0.49, respectively, which remain in close agreement with our findings. Although multiplicity is generally expected to decline over time due to the dynamical loss of companions, the A. L. Kraus et al. (2011) results suggest that, within Taurus, the overall MF remains relatively stable across these early stages of evolution.

Our separation distribution analysis offers further insight into this evolution. As seen in Figure 10, The Taurus+ sample more closely resembles the high-mass subsample from A. L. Kraus et al. (2011) particularly due to the lack of wide companions in the Kraus low-mass group. This is supported by our statistical analysis, which shows that Taurus+ and the Kraus low-mass sample are unlikely to be drawn from the same parent distribution (KS & AD $p = 0.001$), while we cannot reject the null hypothesis for the high-mass comparison ($p_{\mathrm{KS}} = 0.486$ & $p_{\mathrm{AD}} = 0.213$).

Paradoxically, our observed sample is dominated by sources in the low-mass regime (0.25–0.7 $M_\odot$; Coleman-Plante et al., in prep), suggesting that these systems may still be subject to evolutionary processes that shift their distributions toward the unimodal, close-separation peak seen in the Kraus low-mass sample. This evolution would be expected to be driven by two main processes. First, systems that remain bound may undergo inward migration, primarily driven by dynamical friction with the surrounding envelope gas. Second, wide companions, particularly those at separations greater than 1000 au, may become unbound as the system loses mass during early evolution, such as through envelope-clearing outflows or external perturbations (e.g., S. S. R. Offner & H. G. Arce (2014); S. I. Sadavoy & S. W. Stahler (2017)). While D. Guszejnov et al. (2023) suggest that low-mass systems are especially vulnerable to losing wide companions in dense environments due to frequent dynamical interactions, this would be less effective in a low-density region like Taurus. Another possibility for the loss of wide companions is that some may form as unbound systems within the same core. If their relative velocities are sufficiently high and their masses too low, they may never become gravitationally bound.

We also compared our results to the field solar-type binary population from D. Raghavan et al. (2010), which also exhibits a unimodal separation distribution peaking between 30-80 au. Our statistical comparison shows that the extended Taurus+ sample is inconsistent with this distribution ($p_{\mathrm{KS}} = 0.014$, $p_{\mathrm{AD}} = 0.006$), likely due to the excess of wide multiples in Taurus. Given the mass distribution of our sample and the fact that M-type stars are the most common outcome of star formation, many of our sources are likely to evolve into M stars. These stars are observed to have even smaller mean separations ($\sim$20 au; J. G. Winters et al. 2019) further supporting the idea that dynamical evolution will ultimately reduce the wide companion population in Taurus and shift the distribution toward tighter separations.

5.5. *Limitations*

Despite the strengths of our study, several limitations must be acknowledged. First, while our sample is nearly complete for known protostars in Taurus, the region itself contains a relatively small number of young stellar objects, which limits the statistical power of our analysis. Additionally, our reliance on dust continuum and free–free emission to identify companions introduces a selection bias: systems without sufficient circumstellar material or jet activity may go undetected, potentially underestimating the true MF (J. J. Tobin et al. 2016). This bias is mitigated to some extent by the Taurus+ sample, which also includes companions detected in the infrared. A key limitation is the classification of sources into evolutionary stages based on spectral index or bolometric temperature, which can be affected by viewing geometry (B. A. Whitney et al. 2003; A. Crapsi et al. 2008). For instance, edge-on systems may appear more embedded due to obscuration, leading to misclassifications. Some systems previously thought to be Class I may in fact be more evolved but are seen at high inclinations (e.g., IRAS 04260+2642; K. L. Luhman et al. 2010). Furthermore, we lack high-resolution radio data for many sources in the extended sample, leaving open the possibility that unresolved close companions remain undetected. Our angular resolution also limits our ability to detect companions with projected separations smaller than $\sim$18 au. Finally, since we observe each system at a single moment in time and only in two dimensions, the projected separations we report are lower limits; true separations may be larger depending on orbital phase, inclination, and eccentricity (G. P. Kuiper 1935). However, because eccentric systems spend the most time near apastron, we are likely to observe them near their widest separations. These factors should be considered when interpreting the observed multiplicity statistics and separation distributions.

6. CONCLUSIONS

We conducted a high-resolution, multi-wavelength survey using ALMA 0.9 mm and VLA 9 mm continuum observations of 27 protostars in the Taurus molecular cloud to analyze multiplicity, measuring the frequency and separation distribution of multiple systems in this nearby, low-density star-forming region. We extended the sample (Taurus+) by including 24 additional Taurus protostars from the literature not covered in our observations. Using these datasets, we compared multiplicity statistics in Taurus to those in the more clustered Orion and Perseus regions, as well as to a survey of more evolved Class II and III Taurus stars, in order to examine both environmental and evolutionary trends. Our main results are as follows:

1. Within 18–10,000 au separations, the observed sample has an MF of 0.50 and a CF of 0.58, while the Taurus+ sample yields MF = 0.53 and CF = 0.72. These values are higher than those measured in Orion (MF = 0.29, CF = 0.42) and Perseus (MF = 0.36, CF = 0.51). While these differences are only on the $\sim$1–2$\,\sigma$ level, owing to the smaller protostellar population in Taurus, the near completeness of the Taurus+ sample suggests that Taurus may indeed have an intrinsically higher multiplicity rate.

2. The separation distributions in Taurus and Taurus+ (18–10,000 au) both peak near 75 au and show a deficit of companions around 2000 au. The observed sample is sparsely populated beyond 2000 au, whereas Taurus+ fills in some of these wider separations and shows an elevated wide-companion population. Although the small number of protostars in Taurus limits the clarity of these trends, the distributions resemble the more clearly bimodal shapes seen in Orion and Perseus. However, statistical tests indicate that both Taurus ($p_{\mathrm{KS}} = 0.290$, $p_{\mathrm{AD}} = 0.136$) and Taurus+ ($p_{\mathrm{KS}} = 0.176$, $p_{\mathrm{AD}} = 0.403$) cannot be ruled out from having been drawn from a log-flat distribution, so bimodality cannot be confirmed.

3. The elevated multiplicity in Taurus compared to Orion and Perseus may be consistent with predictions from simulations that low-density environments produce higher MFs, experiencing fewer dynamical interactions and thereby preserving primordial multiples (D. Guszejnov et al. 2023).

4. The presence of both close (<200 au) and wide (>1000 au) multiples, together with the pronounced local minimum around 200–300 au in each sample, suggests that both disk and core fragmentation may be needed to reproduce the observed separation distributions, with disk fragmentation primarily forming close multiples and core fragmentation producing the wide multiples.

5. After restricting the Class II/III sample of A. L. Kraus et al. (2011) to the same separation range as ours, their MF (0.56) matches closely with our results ($\sim$0.5), further suggesting that multiplicity in Taurus may be comparatively stable due to the region's more quiescent environment. Within this comparison, we found the Taurus separation distributions more closely resemble their higher-mass subsample, as the low-mass systems show a pronounced lack of wide companions. But, because most protostars in our sample are likely to evolve into low-mass stars (Coleman-Plante et al., in prep.), we expect the separation distribution to change over time, and potentially evolve toward the unimodal distributions observed for the Kraus low-mass sample (peak at 40 au) and the Raghavan field-star population (peak at 30–80 au).

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This research has made use of the SIMBAD database, operated at CDS, Strasbourg, France (M. Wenger et al. 2000).

Facilities: VLA, ALMA

Software: Astropy (http://www.astropy.org; Astropy Collaboration et al. 2013, 2018, 2022), CASA (http://casa.nrao.edu; CASA Team et al. 2022), dustmaps (G. Green 2018)

APPENDIX

A. SOURCE INFORMATION

A.1. *Observed Sources*

IRAS 04016+2610 (L1489 IRS): We classify this source as a Class I protostar, consistent with previous designations in the literature (S. J. Kenyon & L. Hartmann 1995; R. J. White & L. A. Hillenbrand 2004; K. L. Luhman et al. 2010). Our ALMA observations clearly detect a circumstellar disk but reveal no companions. The higher resolution eDisk survey likewise resolved the disk and did not identify any additional sources (N. Ohashi et al. 2023).We incorporated additional photometry from the literature (S. M. Andrews & J. P. Williams 2005; C. H. Young et al. 2003; G. H. Moriarty-Schieven et al. 1994; F. Motte & P. André 2001; J. A. Eisner 2012; M. Barsony & S. J. Kenyon 1992; E. F. Ladd et al. 1991) on top of the VizieR SED for this source.

IRAS 04108+2803 B: Classified as Class I in both K. L. Luhman et al. (2010) and F. Motte & P. André (2001), which our analysis supports. This source has a wide companion observed in previous high-resolution multiplicity studies and is included in the Taurus+ sample ($\sim 0''.33$; M. S. Connelley et al. 2008). In our observations, IRAS 04108+2803-B appears as a single point source. We incorporated additional photometry from the literature (C. H. Young et al. 2003; G. H. Moriarty-Schieven et al. 1994; F. Motte & P. André 2001; M. Barsony & S. J. Kenyon 1992) into the VizieR SED for this source.

IRAS 04158+2805 AB: R. J. White & L. A. Hillenbrand (2004) originally classified this source as a proto-brown dwarf, but K. L. Luhman et al. (2010) later argued that it is instead a low-mass source. Studies have found this source consistent with a Class I protostar (R. J. White & L. A. Hillenbrand 2004; E. Furlan et al. 2008; K. L. Luhman et al. 2010), which our analysis supports. E. Ragusa et al. (2021) reported the first detection of this object as a close binary with a circumbinary disk in ALMA data. Our observations confirm the presence of these close companions and the circumbinary structure. We incorporated additional photometry from the literature (S. M. Andrews & J. P. Williams 2005; F. Motte & P. André 2001) into the VizieR SED for this source.

IRAS 04166+2706: This source has been classified as Class I by R. J. White & L. A. Hillenbrand (2004); K. L. Luhman et al. (2010); F. Motte & P. André (2001), but both our analysis and the eDisk study place it in the Class 0 category (N. Ohashi et al. 2023; N. T. Phuong et al. 2025). In our observations, IRAS 04166+2706 appears as a single point source with no evidence of companions, consistent with the eDisk results, which resolved the disk but did not identify any additional nearby sources. We incorporated additional photometry from the literature (S. M. Andrews & J. P. Williams 2005; C. H. Young et al. 2003; M. Barsony & S. J. Kenyon 1992; J. A. Eisner 2012; F. Motte & P. André 2001) into the VizieR SED for this source.

IRAS 04169+2702: This source has been classified as Class I by R. J. White & L. A. Hillenbrand (2004), K. L. Luhman et al. (2010), and N. Ohashi et al. (2023), consistent with our calculations. In our observations, IRAS 04169+2702 appears as a single point source, with no evidence of companions. M. S. Connelley et al. (2008) reported a close binary with a separation of 28 au; however, this companion is not confirmed by the higher-resolution eDisk study (N. Ohashi et al. 2023). The eDisk team also noted the presence of a cavity, which can be indicative of a close binary. They suggested that such a feature could be caused by a companion even closer than 28 au, though no such companion has yet been directly detected. We incorporated additional photometry from the literature (S. M. Andrews & J. P. Williams 2005; C. H. Young et al. 2003; G. H. Moriarty-Schieven et al. 1994; F. Motte & P. André 2001; J. A. Eisner 2012; M. Barsony & S. J. Kenyon 1992) into the VizieR SED for this source.

IRAS 04181+2654 AB: We detect component A as a point source in both the ALMA and VLA observations, but component B is a non-detection in our data. However, IRAS 04181+2654 B has been reported in other high-resolution surveys (M. S. Connelley et al. 2008) and is therefore included in our extended Taurus+ sample. Both components are consistently classified as Class I sources in the literature (K. L. Luhman et al. 2010; E. Fiorellino et al. 2023; R. J. White & L. A. Hillenbrand 2004), in agreement with our analysis. We incorporated additional photometry from the literature from F. Motte & P. André (2001) for component A, into the VizieR SED for this source.

IRAM 04191 (IRAM 04191+1522) AB: First identified by P. André et al. (1999), IRAM 04191 A was later confirmed as a protostar based on its accretion properties (M.-R. Kim et al. 2016; G. Kim et al. 2019). K. L. Luhman et al. (2010) classified it as a clear Class 0, consistent with our calculations. We detect it as a point source in our Band 7 ALMA data; however, in archival Band 6 ALMA observations we identify a previously unreported companion at a projected separation of ∼1800 au (included in the Taurus+ sample). We associate the pair with the IRAS 04191+1523 AB system. The close companion reported by X. Chen et al. (2012) is not detected in our data.

IRAS 04191+1523 AB: This protostellar binary has been previously reported in the literature (G. Duchêne et al. 2004; M. S. Connelley et al. 2008). Both components are classified as Class I by K. L. Luhman et al. (2010), consistent with our own calculations. IRAS 04191+1523 AB forms an intermediate-separation binary (∼900 au), which we further associate with the wider companion IRAM 04191.

IRAS 04239+2436 AB: Both classified as Class I in both this work and previous studies (R. J. White & L. A. Hillenbrand 2004; K. L. Luhman et al. 2010). We observe two point source objects at a 30 au separation. Previously identified as a close embedded protostellar binary with HST (B. Reipurth et al. 2000), more recent SO line emission observations reveal prominent triple spiral arms surrounding the system, with J.-E. Lee et al. (2023) suggesting it may host three protostars.

IRAS 04248+2612 ABC: A Class I triple system classified as a proto–brown dwarf by (R. J. White & L. A. Hillenbrand 2004) based on its near-IR and optical properties. The system consists of a close binary and a third component at a separation of ∼1500 au previously reported by G. Duchêne et al. (2007) but consistent with our results.

IRAS 04260+2642: This source was initially classified as Class I by R. J. White & L. A. Hillenbrand (2004), but its evolutionary stage remains uncertain because the edge-on disk obscures the stellar light. Our ALMA observations clearly resolve this disk. Some studies have argued that the system may be more evolved than a typical Class I; for example, K. L. Luhman et al. (2010) reclassified it as Class II, noting that while its spectral index falls within the Class I range, the lack of envelope signatures supports a Class II designation. For consistency with our calculated evolutionary classifications, however, we retain a Class I designation. No companions have been reported.

IRAS 04263+2426 (Haro 6-10, GV Tau) AB: This system is a known binary at ∼200 au separation (M. Simon et al. 1995; E. L. Gibb et al. 2007). We observe two compact sources separated by about 175 au. It has been classified as Class I by K. L. Luhman et al. (2010) and R. J. White & L. A. Hillenbrand (2004), which is in agreement with our calculations.

IRAS 04264+2433 AB: This close binary system was first reported by G. Duchêne et al. (2004) in their deep infrared survey, which revealed a faint companion, consistent with our detection. It was previously classified as a Class I source by K. L. Luhman et al. (2010) and R. J. White & L. A. Hillenbrand (2004), consistent with our results.

IRAS 04287+1801 (L1551 IRS5) AB: This is a well-known close binary, previously reported in multiple studies (J. H. Bieging & M. Cohen 1985; L. F. Rodriguez et al. 1995; M. S. Connelley et al. 2008). With ALMA, we observe this 60 au separation binary with a surrounding circumbinary disk. Both companions are classified as Class I in our calculations, which is consistent with previous studies (R. J. White & L. A. Hillenbrand 2004; K. L. Luhman et al. 2010).

L1551NE (IRAS 04288+1802) AB: This source is classified as Class I in our calculations and in K. L. Luhman et al. (2010). Our observations resolve it as a binary with a separation of ∼80 au, with a faint circumbinary disk, consistent with previous work by S. Takakuwa et al. (2014).

IRAS 04295+2251 (L1536 IRS) AB: Classified as a Class I object by K. L. Luhman et al. (2010) and R. J. White & L. A. Hillenbrand (2004), from our calculations we likewise adopt a Class I designation. Our ALMA observations show a clear circumstellar disk with a large central cavity, and we present sub-arcsecond resolution 9 mm archival VLA observations that show a potential close companion (18 au separation) that we include in the Taurus+ sample. We incorporated additional photometry from the literature (S. M. Andrews & J. P. Williams 2005; C. H. Young et al. 2003; G. H. Moriarty-Schieven et al. 1994; F. Motte & P. André 2001; J. A. Eisner 2012) into the VizieR SED for this source.

IRAS 04302+2247: Classified as a Class I source by K. L. Luhman et al. (2010), we also adopt this Class I designation. We observe an edge on disk around this single source, which could serve to obscure the central star and make its evolutionary stage uncertain. This edge on disk was also observed in the eDisk survey (N. Ohashi et al. 2023) and by S. Wolf et al. (2008) with no additional companions reported. We incorporated additional photometry from the literature (S. M. Andrews & J. P. Williams 2005; C. H. Young et al. 2003; G. H. Moriarty-Schieven et al. 1994; F. Motte & P. André 2001) into the VizieR SED for this source.

IRAS 04325+2402 AB: This source is an intermediate binary (∼1000 au) with two components. It was initially suggested to be a triple system, consisting of a close binary with a wide companion (L. Hartmann et al. 1999), but later work showed that the proposed close binary is a single protostar with an absorption lane along the line of sight (A. Scholz et al. 2010). This is a Class I system according to our analysis, which is consistent with previous classifications from the literature (R. J. White & L. A. Hillenbrand 2004; K. L. Luhman et al. 2010). With ALMA we observe two resolved disks with orthogonal orientations.

IRAS 04361+2547 (TMR-1) AB: Classified as Class I in both this study and K. L. Luhman et al. (2010). Hubble Space Telescope (HST) imaging by S. Terebey et al. (1998) revealed a binary with a separation of ∼30 au. In our data, we detect only a single source, but include IRAS 04361+2547 B in the Taurus+ sample given its confirmation in high-resolution observations.

IRAS 04365+2535 (TMC 1A): Classified as Class I by R. J. White & L. A. Hillenbrand (2004) and K. L. Luhman et al. (2010); our analysis supports the same classification. We detect it as a single point source, in line with previous high-resolution surveys (G. Duchêne et al. 2007; M. S. Connelley et al. 2008). Later, P. Bjerkeli et al. (2016) used ALMA to resolve structure down to 6 au and also found no companions. We incorporated additional photometry from the literature (S. M. Andrews & J. P. Williams 2005; G. H. Moriarty-Schieven et al. 1994; F. Motte & P. André 2001; J. A. Eisner 2012; N. Ohashi et al. 1996; C. J. Chandler & J. S. Richer 2000) into the VizieR SED for this source.

IRAS 04368+2557 (L1527 IRS): The source is widely recognized as a Class 0 object, consistent with its low bolometric temperature ($T_{\rm bol} < 70$ K, H. Chen et al. (1995); F. Motte & P. André (2001); K. L. Luhman et al. (2010)). L. Loinard et al. (2002) previously reported it as a binary based on VLA 7 mm observations with a separation of 25 au, but R. Nakatani et al. (2020) and P. D. Sheehan et al. (2022) later confirmed it as a single star and noted potential substructure in the disk. In our data, we also detect it as single source and observe a resolved edge on disk. The eDisk study resolved a similar edge on disk and no companions for this source (N. Ohashi et al. 2023).

IRAS 04381+2540 (TMC-1, L1534) AB: This source was first published as a binary after being detected with HST by D. Apai et al. (2005). With ALMA and VLA we observe two compact sources at a separation of 85 au. We designate it as a Class I source, which is consistent with the classification in the literature (R. J. White & L. A. Hillenbrand 2004; K. L. Luhman et al. 2010).

IRAS 04385+2550 (Haro 6-33): Initially classified as a Class I source by R. J. White & L. A. Hillenbrand (2004), K. L. Luhman et al. (2010) reclassified it as Class II, noting that although its spectral index falls within the Class I range, the absence of envelope signatures supports a Class II designation. For consistency with our calculated evolutionary classifications, we still consider this a Class I source. We observe a single point like source and no companions have been previously reported.

IRAS 04489+3042 AB: Initially classified as a proto–brown dwarf by (R. J. White & L. A. Hillenbrand 2004), later reinterpreted by K. L. Luhman et al. (2006) as a low-mass star. Our observations identify it as a Class I binary with a faint companion.

DG Tau (IRAS 04240+2559) AB: This is a well-known wide binary system. (R. J. White & L. A. Hillenbrand 2004) classified both components as Class II, while K. L. Luhman et al. (2010) classified DG Tau B as Class I and DG Tau A as Class II. In our analysis, we calculate DG Tau B as Class I and DG Tau A as FS. With both ALMA and VLA observations centered on DG Tau B, DG Tau A is only detected in the VLA data, owing to its wider field of view. DG Tau B shows a resolved disk in our ALMA observations, and both components appear as compact point sources in the VLA data. previous high-resolution ALMA observations reveal a circumstellar disk around DG Tau A as well (M. Yamaguchi et al. 2024).

HH 30: Classified as Class I by K. L. Luhman et al. (2010), this source hosts a nearly edge-on disk that obscures the central star, and creates ambiguity for its evolutionary stage. Other studies suggest it may be more evolved (R. J. White & L. A. Hillenbrand 2004). We adopt a Class II designation based on our own calculations. I. Joncour et al. (2017) associated this object with LkHα358, but their reported separation of ∼11,350 au is beyond the separation limit considered in our study. From our data, we measure this source as a single disk.

A.2. *Extended Taurus+ Sources*

IRAS 04108+2803-A: A wide companion (∼2500 au) to 04108+2803 B, though undetected in our observations of that source. We include it in the extended sample as it has been reported as a companion in previous high-resolution multiplicity studies (∼0.″33; M. S. Connelley et al. 2008). This source is classified as Class I in K. L. Luhman et al. (2010) and as Class II in F. Motte & P. André (2001), but we designate it a FS source.

IRAS 04154+2823: This source was originally classified as Class II by S. J. Kenyon & L. Hartmann (1995), but later studies by E. Furlan et al. (2008) and S. M. Andrews & J. P. Williams (2005) reclassified it as a FS source, noting that its SED appears transitional between Class I and Class II. We also have designated it a FS source. No high-resolution IR or radio observations have yet identified any close companions.

IRAS 04166+2708: K. L. Luhman et al. (2010) and R. J. White & L. A. Hillenbrand (2004) classified this source as Class I, which is consistent with our calculations. This source has no known companions but has not yet been observed at sub-arcsecond resolution at IR or radio wavelengths.

2MASS J04194657+2712552 ([GKH94: 41)] K. L. Luhman et al. (2010) classified this source as "Class I?", while E. Furlan et al. (2011) found it to be disk-dominated and likely more evolved. Our analysis yields a FS classification. It has no known companions but has never been observed at sub-arcsecond resolution at IR or radio wavelengths.

IRAS 04181+2655: Located near the IRAS 04181+2654 system but outside the 10,000 au limit to be considered a companion. Classified as Class I by R. J. White & L. A. Hillenbrand (2004) and as Class II by E. Fiorellino et al. (2023) and K. L. Luhman et al. (2010), though the latter note that while its spectral index is consistent with a Class I source, it lacks evidence for an envelope. For consistency, we adopt the Class I classification. The IRAS designation refers to a composite position that encompasses multiple nearby sources, and as a result the SIMBAD entry for this source is offset, the correct counterpart is 2MASS J04210795+2702204. Even so, we adopt the IRAS name for consistency with the literature.

FS Tau (Haro 6-5, IRAS 04189+2650) AB: Identified as a wide binary with a circumbinary disk in the ALMA Band 6 multiplicity survey of R. L. Akeson et al. (2019) (25–30″resolution). FS Tau A is a spectroscopic binary (P. Hartigan & S. J. Kenyon 2003), but we do not include this in our sample to stay consistent with our resolution limits. While classified as Class I by E. Fiorellino et al. (2023) and K. L. Luhman et al. (2010), and as Class II by R. L. Akeson et al. (2019), our analysis designates FS Tau A as FS and FS Tau B as Class I.

L1521F: This source is deeply embedded within one of the densest cores in Taurus. High-resolution ALMA Band 7 (0.87 mm) continuum observations at 25 au resolution revealed compact emission (K. Tokuda et al. 2017). More recent 1.3 mm observations at 4 au resolution detected spike-like structures extending form the source, but were interpreted as signatures of magnetic flux transport rather than fragmentation K. Tokuda et al. (2024). To date, there is no evidence for true protostar companions, and we classify this source as Class 0.

2MASS J04293209+2430597: This source was listed as "Class I?" by K. L. Luhman et al. (2010), which our analysis likewise supports, so we adopt a Class I designation. No companions are currently known, though it has not been observed at sub-arcsecond resolution at IR or radio wavelengths.

LkHα358: While a previous high resolution study by A. L. Wallace et al. (2020) ($\sim$0.″2) classifies this as a single source, our analysis associates it with HL Tau and XZ Tau at a wide separation of $\sim$6700 au. K. L. Luhman et al. (2010) designates LkHα358 as a Class I source, but our calculations place it in the Flat Spectrum category.

HL Tau (Haro 6-14): A Class I protostar with a well-studied protoplanetary disk exhibiting extensive substructure (ALMA Partnership et al. 2015). It is a wide companion to XZ Tau (Haro 6-15), a Class II spectroscopic binary (P. Hartigan & S. J. Kenyon 2003). The evolutionary classifications for both systems were determined by K. L. Luhman et al. (2010) and are confirmed by our analysis.

Haro 6-13: Listed as Class II by K. L. Luhman et al. (2010) and R. J. White & L. A. Hillenbrand (2004), but reported in the Class I/FS sample of C. Flores et al. (2024). Our analysis classifies it as a FS source. High-resolution studies (C. Leinert et al. 1993; M. S. Connelley et al. 2008) confirmed it as single, and sub-arcsecond resolution ALMA Band 6 data from F. Long et al. (2019) likewise showed no evidence of companions.

HP Tau (IRAS 04328+2248): A protostar that is widely separated from three known companions: HP Tau G2, HP Tau G3-A, and HP Tau G3 B (M. Simon et al. 1995; R. J. Harris et al. 2012; A. C. Rizzuto et al. 2020). HP Tau G3 A and B form a spectroscopic binary and are treated as a single source in our analysis to remain consistent with our resolution limits. High-resolution ALMA Band 6 data from F. Long et al. (2019) showed no companions within their FOV, as expected. We classify HP Tau as Flat Spectrum, while assigning Class II to HP Tau G2 and G3. Although S. M. Andrews & J. P. Williams (2005) also reported HP Tau as Flat Spectrum, K. L. Luhman et al. (2010) classified it as Class II and its companions as Class III.

Haro 6-28 (V1026 Tau) AB: This close pair (∼100 au separation) was first published by C. Leinert et al. (1993) and then by M. Simon et al. (1995). In our analysis it is part of a larger multiple system with five components. C. Flores et al. (2024) classified it as Class I/FS, whereas K. L. Luhman et al. (2010) classified it as Class II. From our calculations we designate it a Class II source.

IRAS S04361+2331 (2MASS J04390525+2337450): Previously classified as "Class I?" by K. L. Luhman et al. (2010), but our analysis classifies it as a FS source. No companions are currently known, though it has not been observed at sub-arcsecond resolution at IR or radio wavelengths.

IC 2087 IR (IRAS 0437+257P08): This source was classified as Class II by R. J. White & L. A. Hillenbrand (2004) but as Class I by K. L. Luhman et al. (2010). From our calculations we assign it a Class I designation. High-resolution observations (∼0.″33) by M. S. Connelley et al. (2008) indicate that it is a single source.

B. MODELED SOURCE SEDS

To establish evolutionary classes and calculate bolometric quantities for each target, we assembled an SED based on archival infrared photometry using the VizieR SED tool (F. Ochsenbein et al. 2000). We used query radii ranging from 4″ to 20″, depending on nearby sources and the available photometry, centered on the coordinates listed in Tables 3 and 4. We then binned the remaining photometry by wavelength, taking the median flux within each bin and using the sample standard deviation as an uncertainty. We removed the 12.663×10^3 GHz point from the Spitzer/MIPS 24 μm instrument because it appeared anomalously high across most source SEDs. For some sources, we included additional photometry from the literature (see Appendix A for references). In this section, we present the SEDs of each source. To accurately show uncertainties on logarithmic axes, we used log-symmetric error bars rather than linear symmetric ones. These are derived from the fractional (or relative) flux uncertainties given by $r = \sigma_F/F$. In Table 8, we present the photometry used to construct the SEDs, including binned data from VizieR and additional measurements from the literature. This table also lists the query radius used for each source in our VizieR searches.

We fit the SED as the sum of (i) a smooth continuum represented by a 6th-order polynomial in log space over the observed range and (ii) a modified blackbody (isothermal $T = 30$ K, $\kappa_\lambda \propto \lambda^{-1.8}$) to extend the SED from the longest observed wavelength to 1000 μm when necessary; the modified blackbody was scaled to match the observed SED. Bolometric luminosities were computed from

$$L_{\rm bol} = 4\pi D^2 \int F_\nu \, d\nu, \tag{B1}$$

and bolometric temperatures from

$$T_{\rm bol} = 1.25 \times 10^{-11} \, \langle \nu \rangle \ {\rm K}, \tag{B2}$$

where

$$\langle \nu \rangle = \frac{\int \nu F_\nu \, d\nu}{\int F_\nu \, d\nu}. \tag{B3}$$

We calculated the infrared spectral index α between 2 and 24 μm from

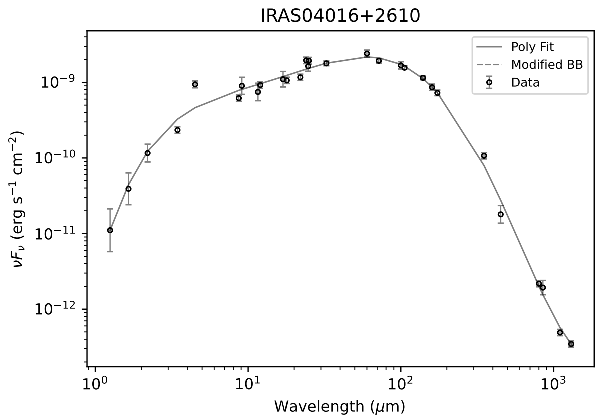
$$\alpha = \frac{\Delta \log(\nu F_\nu)}{\Delta \log \nu}, \tag{B4}$$

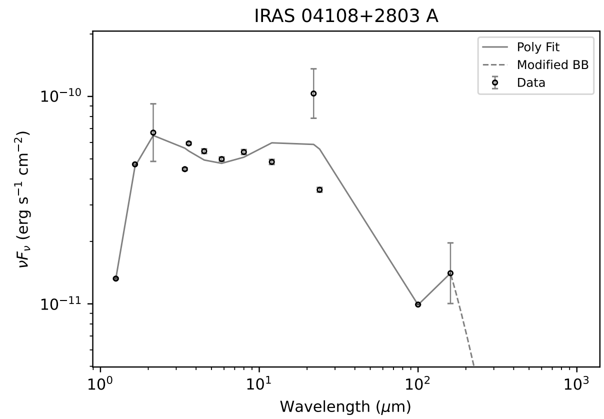
where we used the binned SED points closest to those wavelengths, and adopted the standard class boundaries: $\alpha \geq 0.3$ (Class I), $-0.3 \leq \alpha < 0.3$ (Flat), $-1.6 \leq \alpha < -0.3$ (Class II), and $\alpha < -1.6$ (Class III). We primarily use the infrared spectral index to assign evolutionary classes, but because Class 0 protostars are defined exclusively by their bolometric temperature ($T_{\rm bol} \leqslant 70$ K), we use $T_{\rm bol}$ to classify the four Class 0 sources in our sample (H. Chen et al. 1995). In most cases, they agree. The resulting values and classifications are listed in Tables 3 and 4.

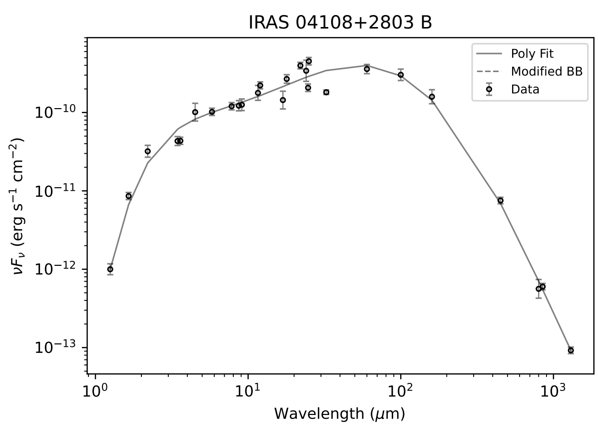
The photometry is primarily based on mid- to far-infrared observations from instruments such as Spitzer, WISE, and Herschel, which have angular resolutions $\gtrsim 1''$. At the distance of Taurus (140 pc), these telescopes cannot resolve close multiples with separations ∼200-1000 au, meaning many of our high-resolution ALMA and VLA detections appear as unresolved single sources in the mid- to far-infrared.

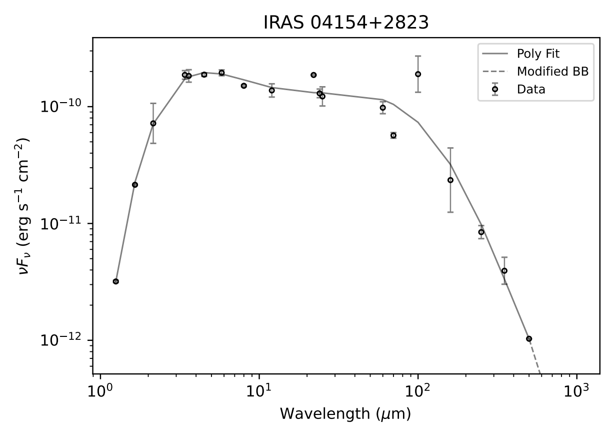
As a result, all close binary systems in our sample, and even some wide multiple systems inherit their $L_{\rm bol}$, $T_{\rm bol}$, and classification from the blended infrared source. In most cases, we were able to resolve individual SEDs for companions with separations greater than 1000 au to get reliable individual $L_{\rm bol}$ and $T_{\rm bol}$ estimates, with the exception of IRAS 04248+2612 ABC at separation of 1647 au (∼12″).

The following figures show the SEDs for all sources in our samples. Each SED includes the measured photometry (points) and the corresponding polynomial fit and modified blackbody model (when necessary). The large error bars associated with some photometry reflect a large scatter in the measurements available in the literature that can result from differing sensitivity, aperture sizes, and methods.









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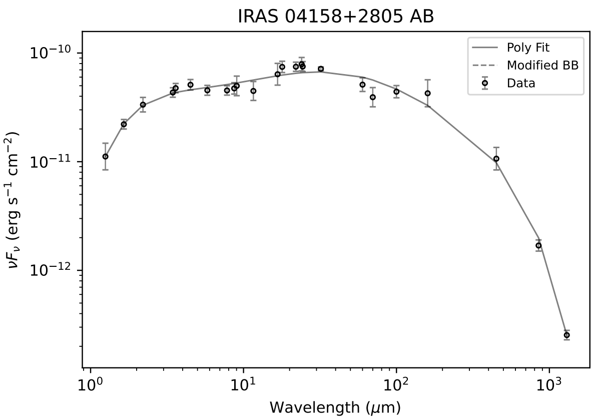

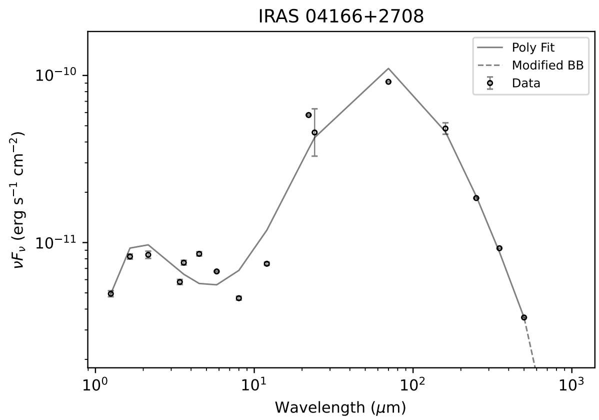

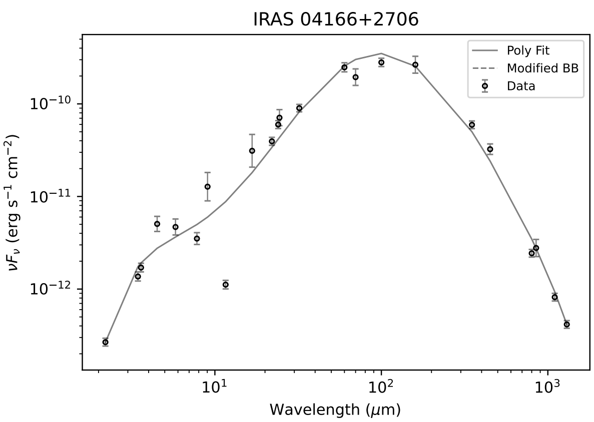

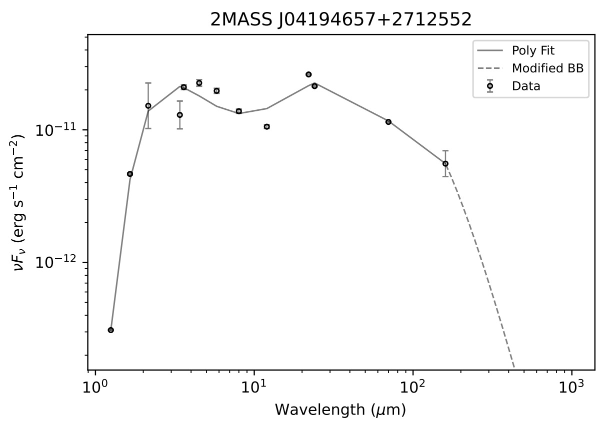

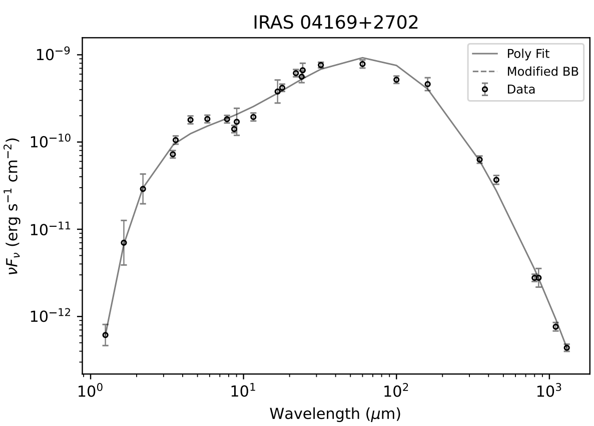

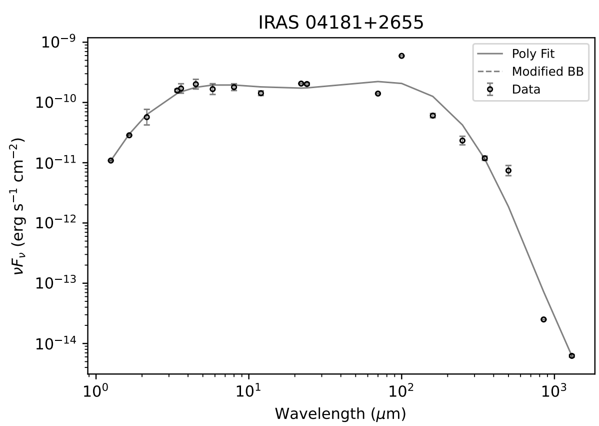

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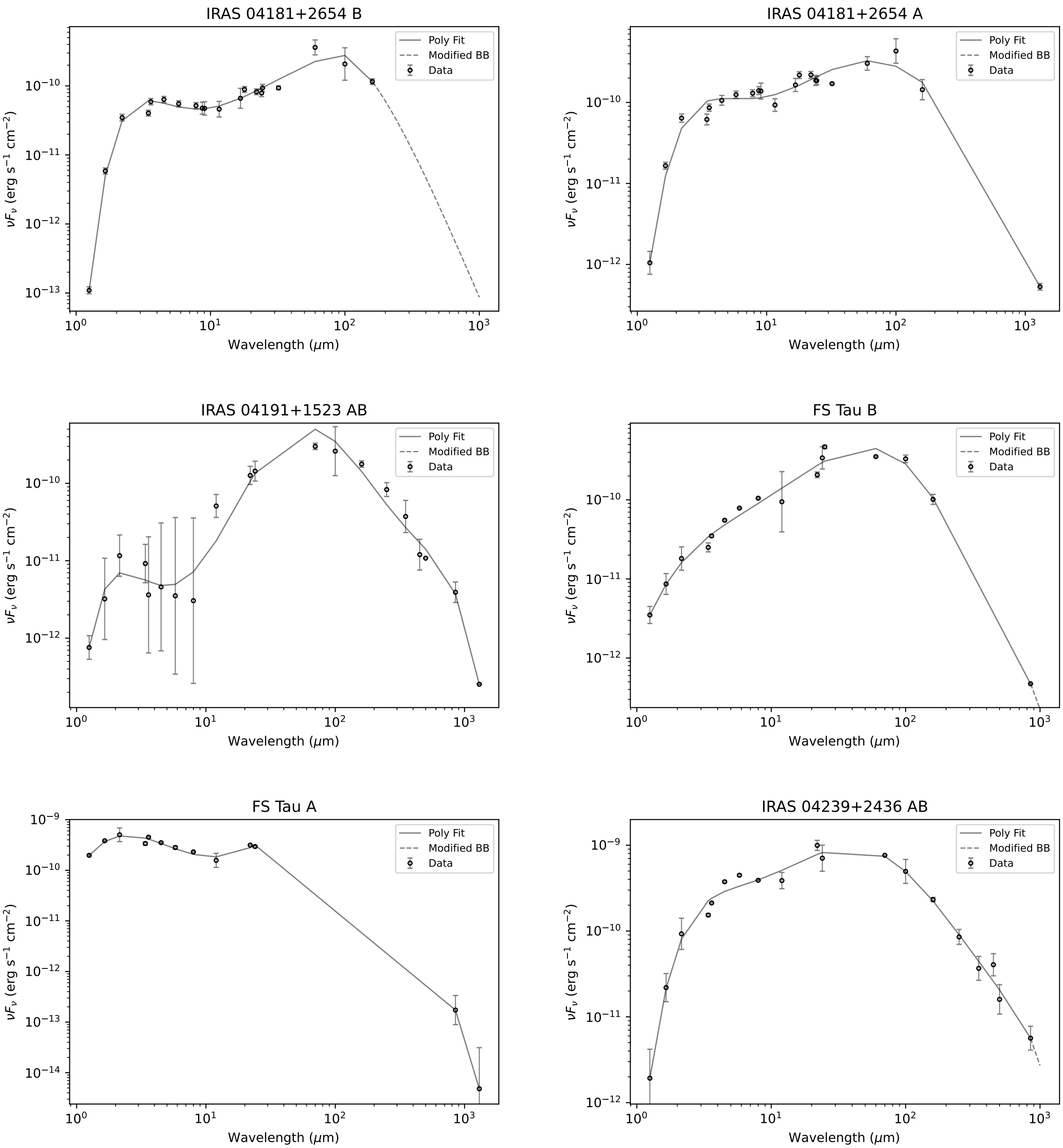

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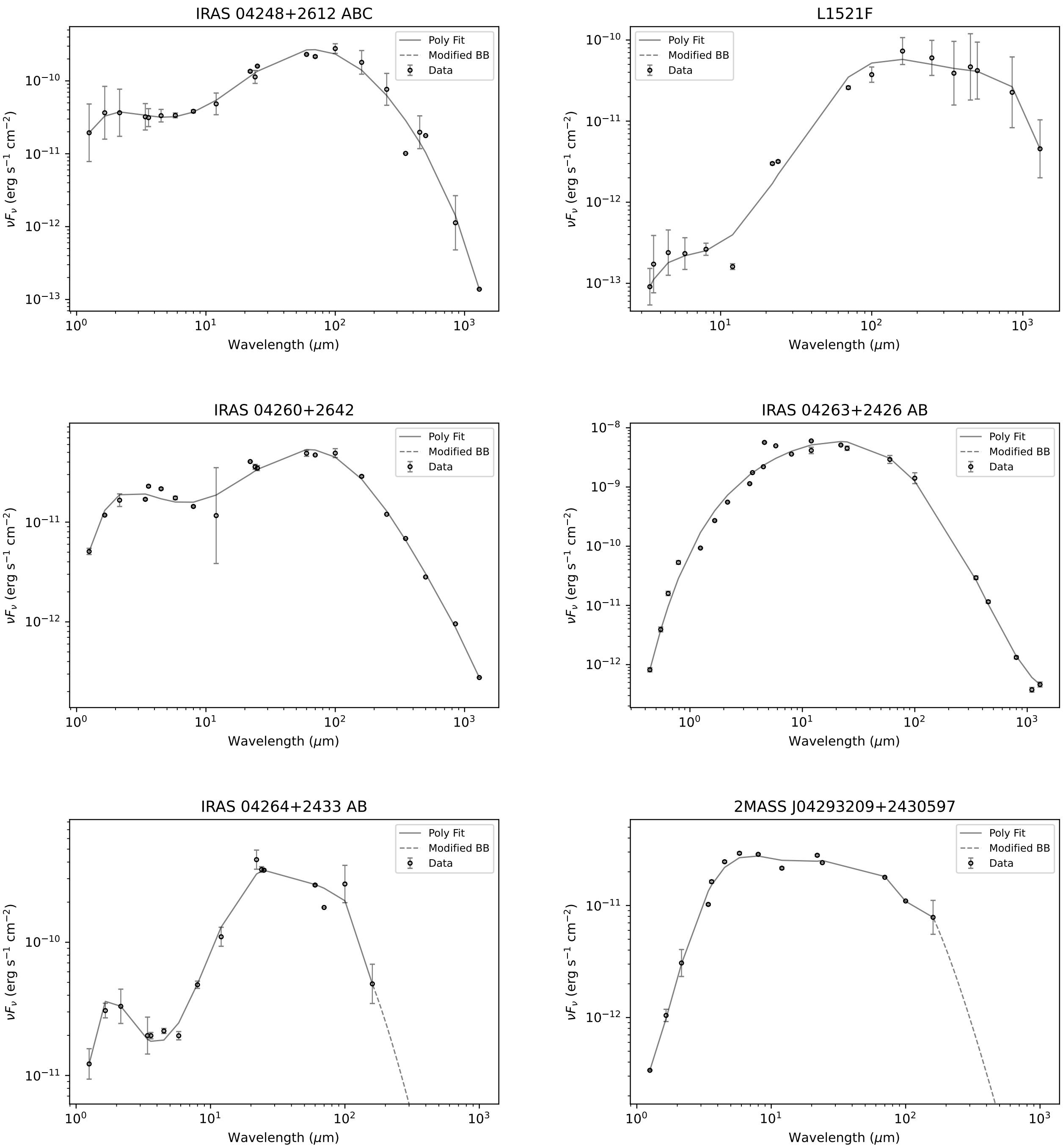

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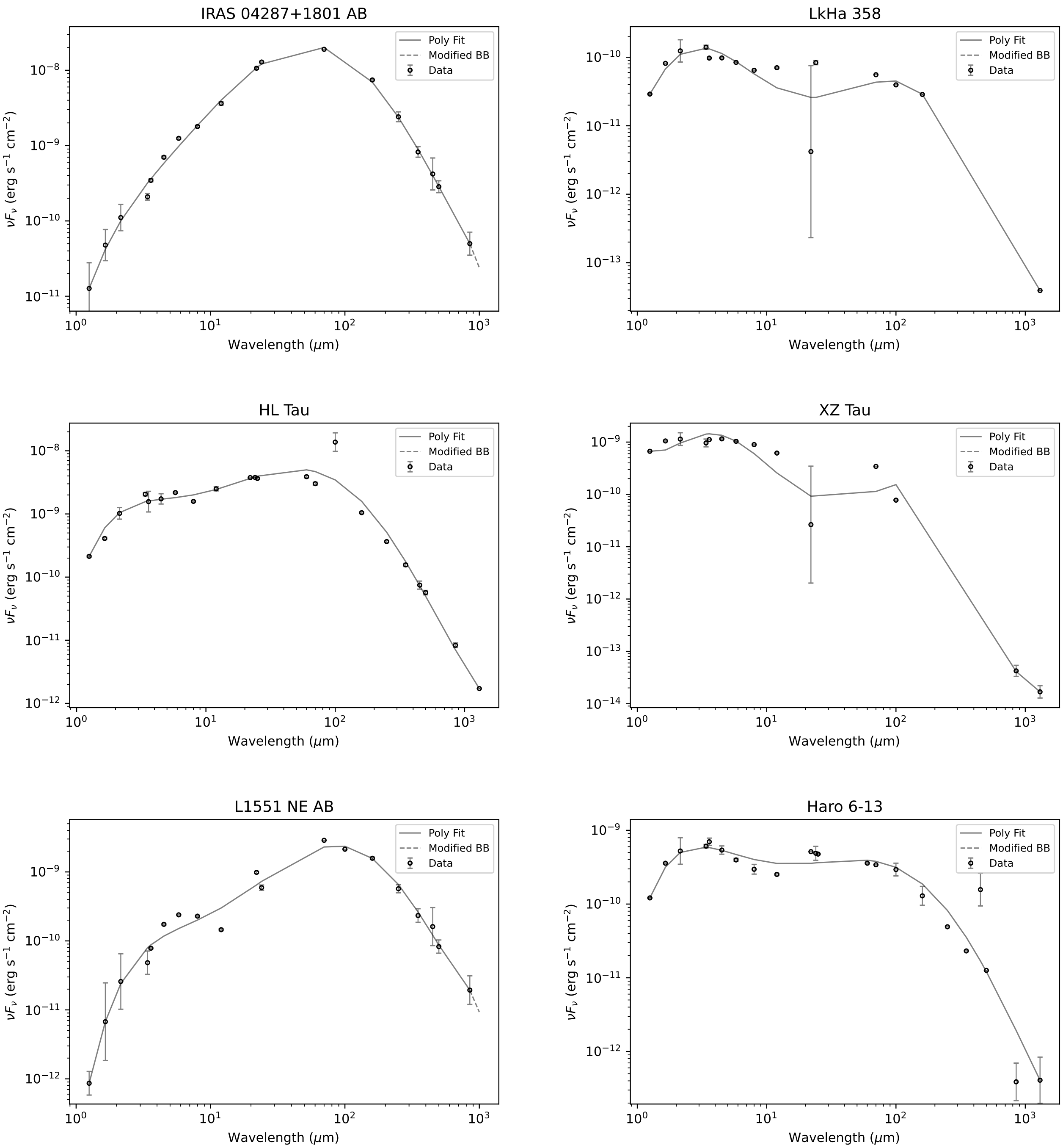

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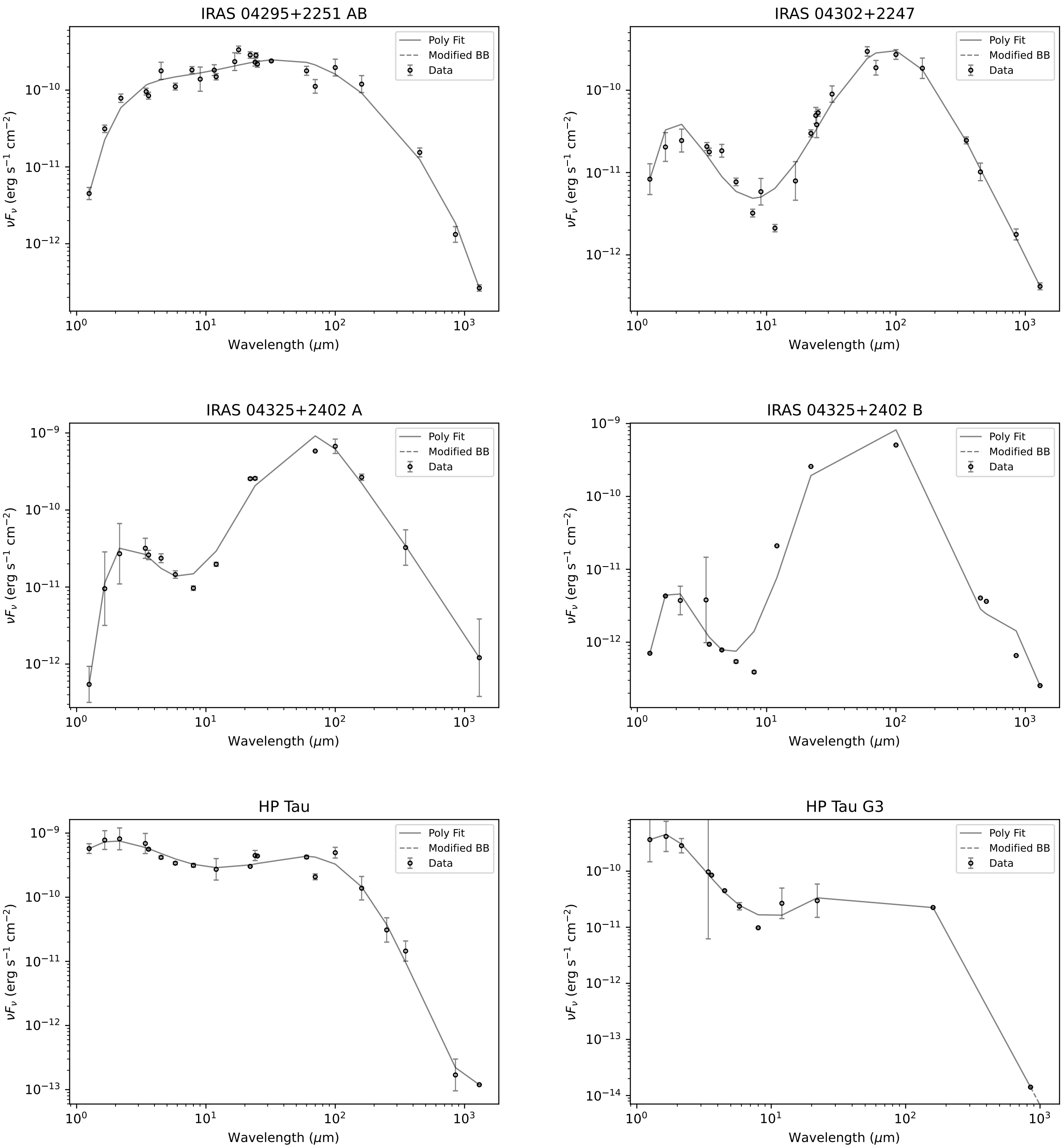

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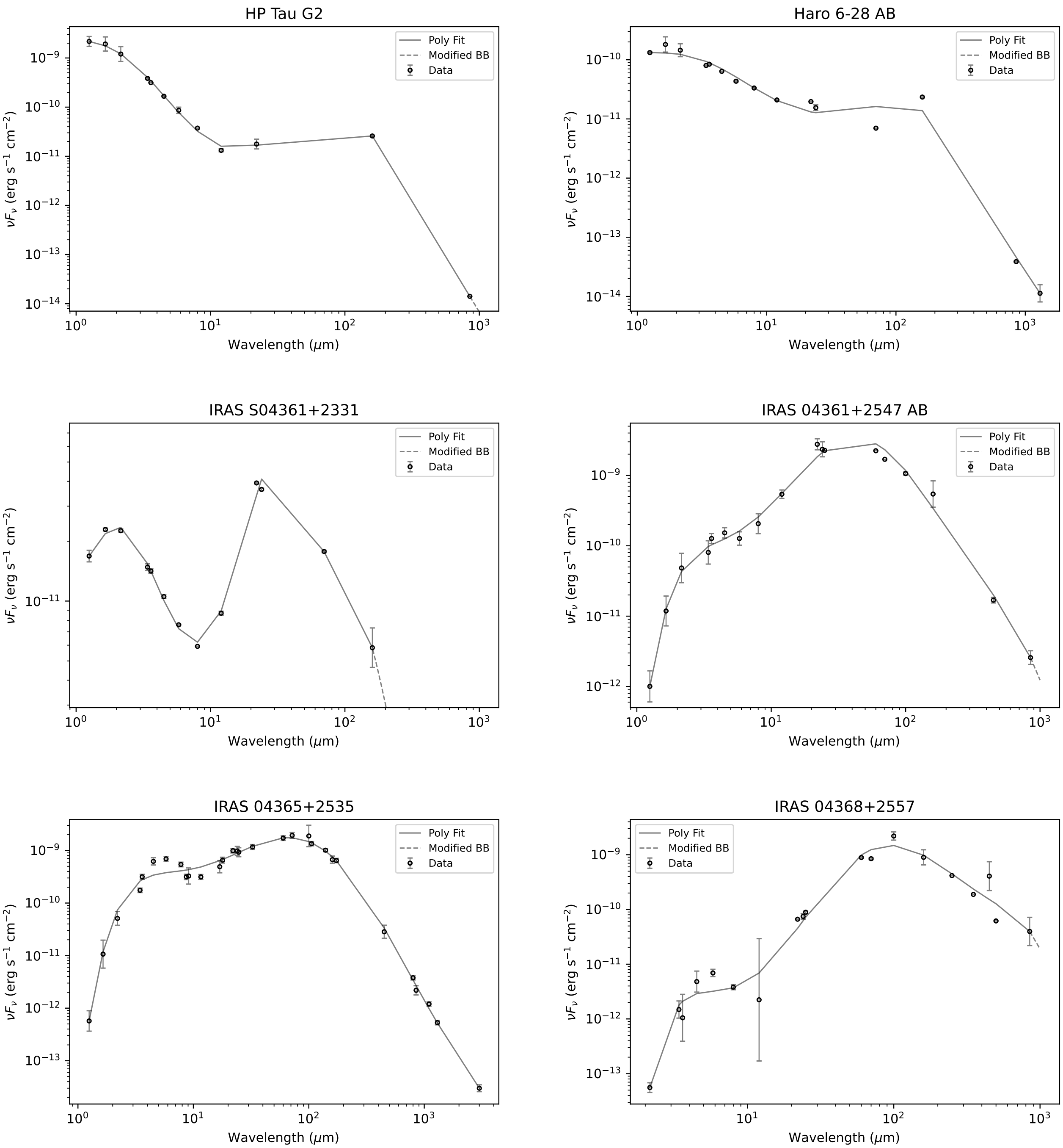

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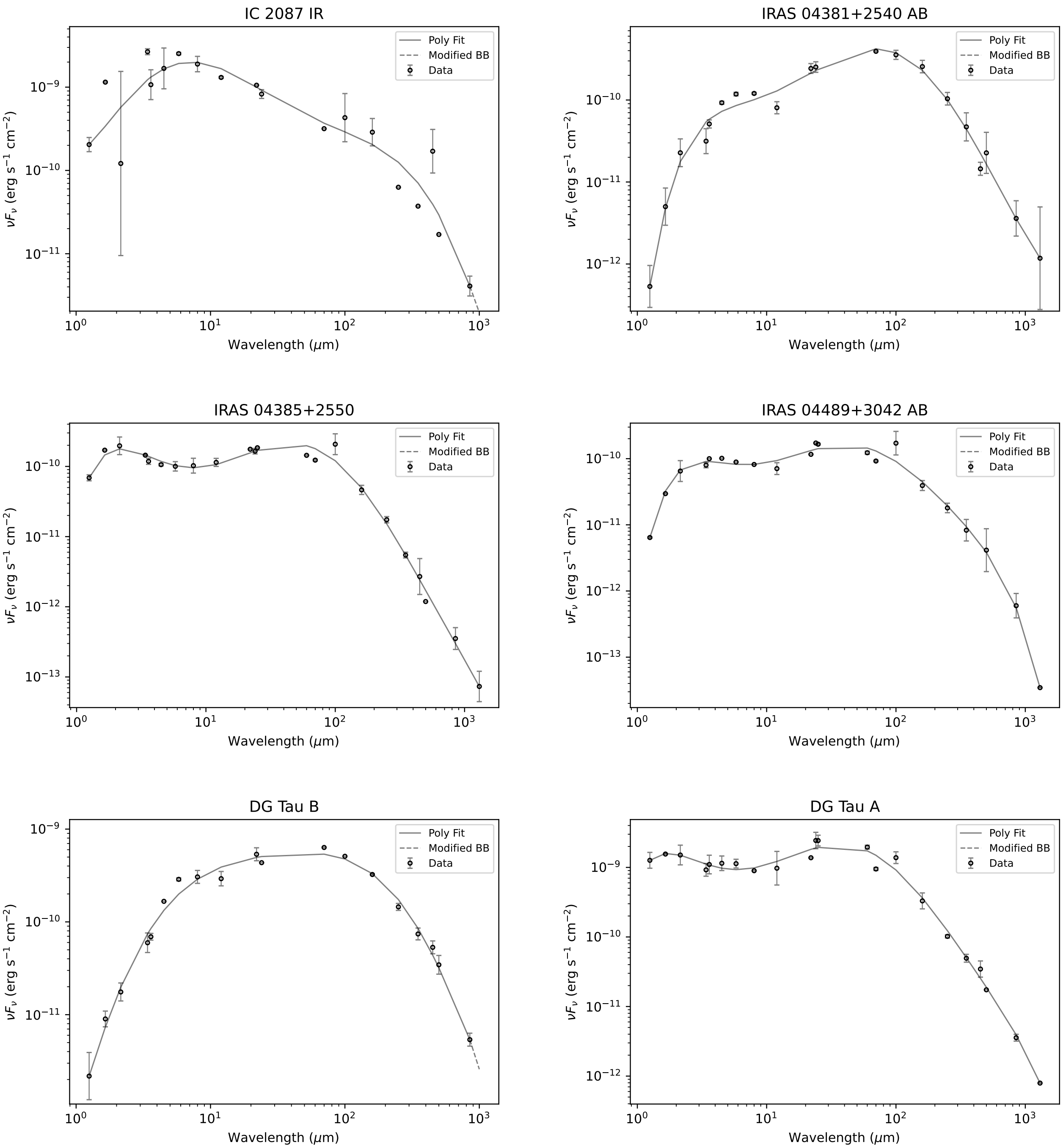

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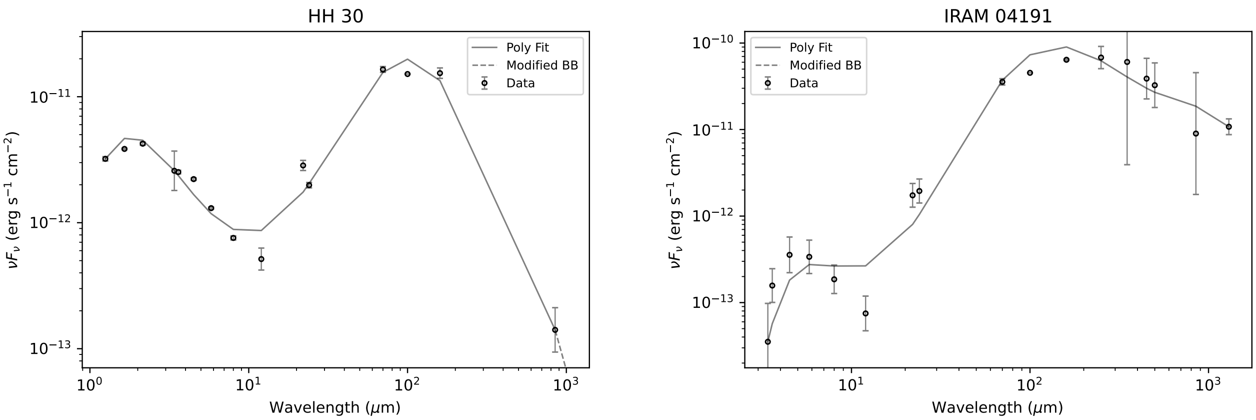

Table 8. SED Photometry

Row number	Source	Query radius (″)	1.25	1.65	2.15
1	IRAS 04016+2610	10″	0.0046	0.0215	···
2	IRAS 04108+2803 B	10″	0.0004	0.0047	···
3	IRAS 04158+2805 AB	10″	0.0046	0.0121	···
...	...	...	...	...	...
53	IC 2087 IR	10″	0.0852	0.632	0.0869

Note—Wavlengths are un units on μm and the corresponding flux values in units of Jy. This table is published in its entirety in the machine-readable format. A portion is shown here for guidance regarding its form and content.

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